Week - 02
Image Processing & Open Source tools

Based on Ellen Eyth Slides

images - what is dpi, ppi & resolution

* **DPI**, or dots per inch is the terminology used when describing the resolution of a printed image. It indicates the number of dots in a printed inch. The image is of higher quality (in sharpness and detail), if the image has a high dpi value. This term is often misused, as when someone asks for a 300dpi quality image or graphic, they usually mean 300ppi.
* **PPI**, or pixels per inch is the terminology used to describe the pixel density of an image on the screen, as opposed to the dpi of a printed image or graphic.
* **Image resolution** refers to the number of pixels in an image, or the detail it holds. A pixel is just one unit of the whole digital image and the resolution depends on the size of the pixels it contains. The higher the resolution, the more detail. Resolution can also refer to the width and height of an image as well as the total number of pixels.

http://www.theprintgroup.com.au/information-bank/images-what-is-dpi-ppi-resolution.html

PPI example

* For example:
  - Image 1 is 297 x 210mm (A4) and 300ppi.
  - Image 2 is 50 x 35mm and 300ppi.
* While both images offer the same ppi resolution, the width and height of each image is different. Each image will only remain at 300ppi at its original size. The ppi resolution of each image will change depending on if you are making the file smaller or larger. If you try to increase the width and height of image 2 to fit a 297 x 210mm (A4) area, you would effectively be reducing the image quality to 50ppi and dramatically reducing sharpness and detail of your printed material.
* So when you are asked for a high resolution, 300ppi image, it is best to purchase or supply, the largest size (in height and width) available. This will make it much easier to resize without effecting the ppi resolution and quality of the print.
* http://www.theprintgroup.com.au/information-bank/images-what-is-dpi-ppi-resolution.html

DETAIL OF IMAGE 1
original size 297 x 210mm (A4) @ 300ppi

DETAIL OF IMAGE 2
original size 50 x 35mm @ 300ppi
enlarged to 297 x 210mm (A4) - now 50ppi

Definitions & Basics of Digital Image

1. Image
2. Digital Image
3. Raster
4. Vector
5. Image Editing

Basic Terminology

The Image*

An image: is an artifact, that has a similar appearance to some subject - usually a physical object or a person.

2-D such as a photograph, screen display, and

3-D such as a statue or hologram.

Basic Terminology

An digital image:

Basic Terminology

Raster Images.

Raster images have a finite set of digital values, called picture elements or pixels.

The digital image contains a fixed number of rows and columns of pixels.

Pixels are the smallest individual element in an image, holding quantized values that represent the brightness of a given color at any specific point.

Basic Terminology

Raster Images.

Raster image type

1. **Binary**: is a digital image that has only two possible values for each pixel.
2. **Grayscale**: is an image in which the value of each pixel is a single sample, that is, it carries only intensity information.
3. **Color**: is a digital image that includes color information for each pixel.

Basic Terminology

<u>Vector Images</u>.

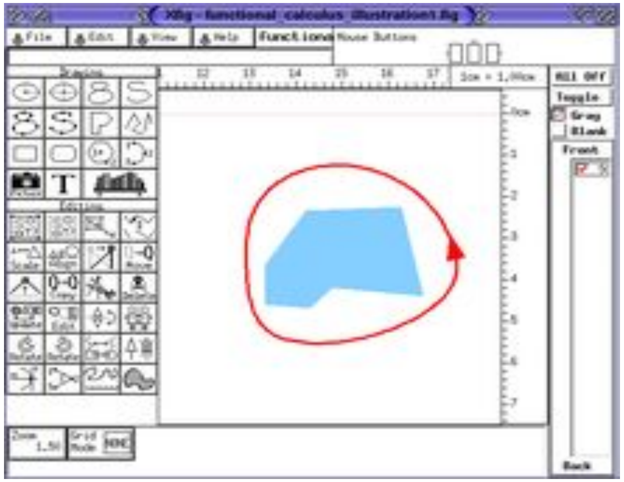
Vector images resulted from mathematical geometry (vector). In mathematical terms, a vector consists of point that has both direction and length.

Often, both raster and vector elements will be combined in one image; for example, in the case of a billboard with text (vector) and photographs (raster).

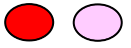
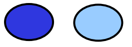
Basic Terminology

Image Editing*

Graphic Software programs, which can be broadly grouped into vector graphics editors,



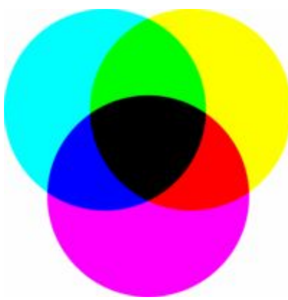
Color Terms

1. ***Hue*** distinguishes among colors (e.g., red, green, purple, and yellow)
2. ***Saturation*** refers to how pure the color is, how much **gray** is mixed with it
    - red saturated; pink unsaturated 
    - royal blue saturated; sky blue unsaturated 
    - Pastels are less saturated, hence less vivid and less intense
3. ***Lightness***: perceived achromatic intensity of reflecting object
4. ***Brightness***: perceived intensity of a self-luminous object, such as a light bulb, the sun, or an LCD screen

Color Mixture

- The effect of (A) passing light through several filters (subtractive mixture), and (B) throwing different lights upon the same spot (additive mixture)

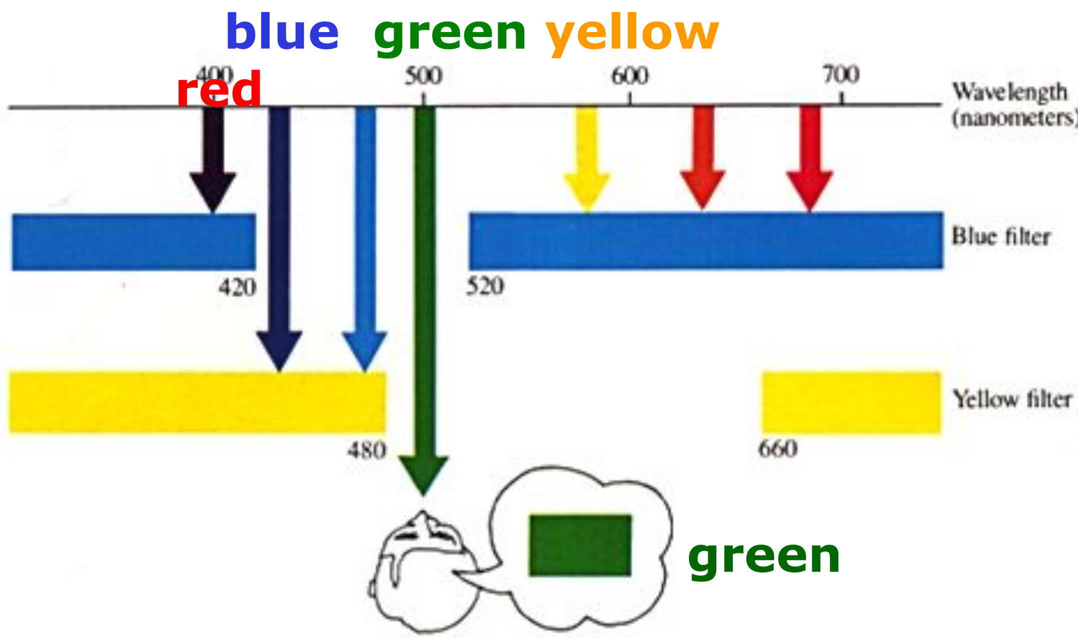
A



B

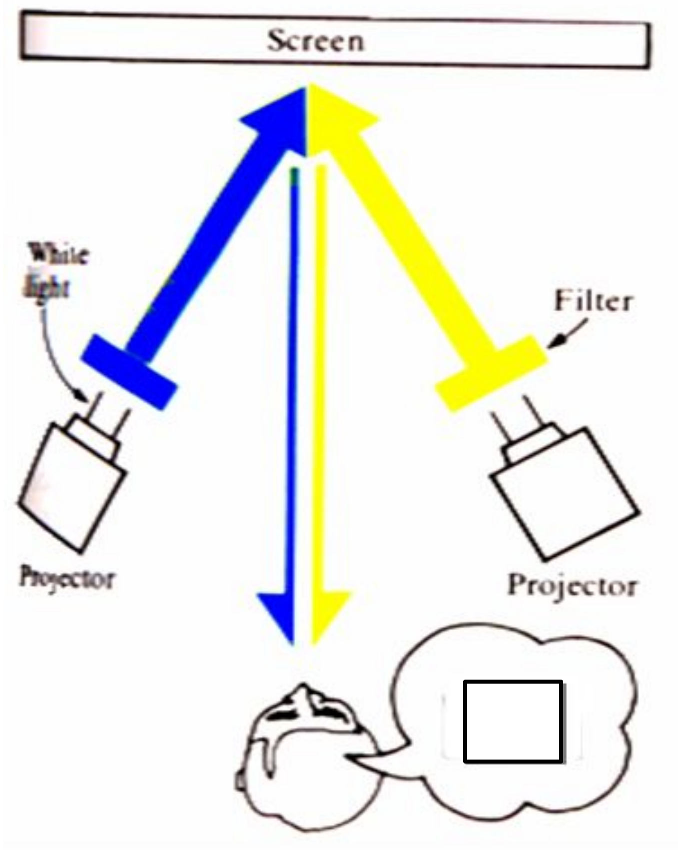
1-Subtractive Mixture الطباعة بالأحبار

- *Subtractive mixture* occurs when mixing paints, dyes, inks, etc. that act as a filters between the viewer and the light source / reflective surface.
- In subtractive mixing, the light passed by two filters (or reflected by two mixed pigments) are wavelengths that are passed by the two filters



2-Additive Mixture للشاشات

- *Additive mixture* occurs when color is created by mixing visible light emitted from light sources
  - Used for computer monitors, televisions, etc.
- Light passed by two filters (or reflected by two pigments) impinges upon same region of retina
- On the diagram: pure blue and yellow (green+red) filtered light on same portion of the screen, reflected upon same retinal region, produce white/gray



COLOR MODELS

- Color Model (Color Space, Color System)

  -Specify colors in a standard way

  -A coordinate system that each color is represented by a single point.

- Most used models:

  -RGB model (Monitor/TV)

  -CYM model (3-color Printers)

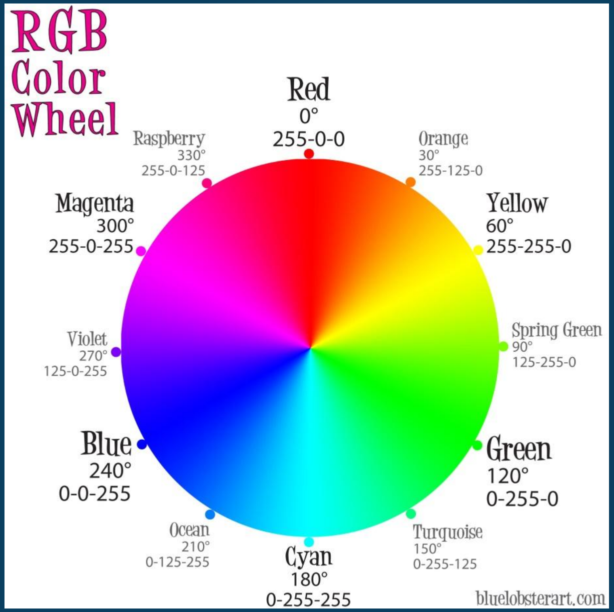
  -CYMK model (4-color Printers)

  -HSI model (Color Image Processing and Description)

Color Models

1- RGB

2- CMYK



1-Color Models - RGB

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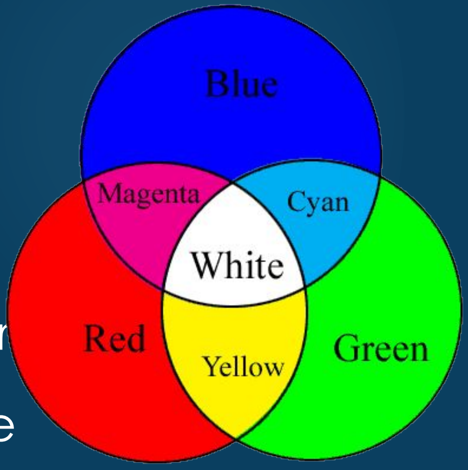
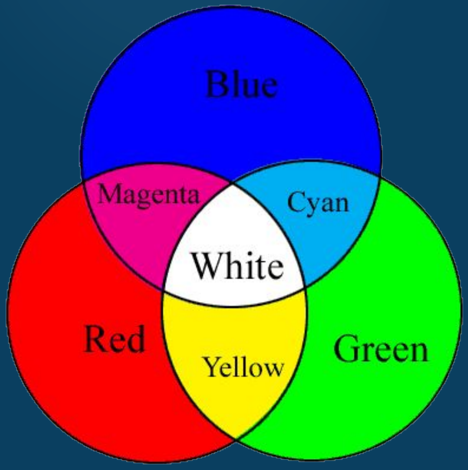
- Additive Color: RGB
  - Describes colors that emanate from glowing bodies such as lights, TV, and computer monitors
  - In additive color models, mixing two colors results in a brighter color
- **Additive Colors**
  are created by mixing spectral light in varying combinations. The most common examples of this ar television screens and computer monitors, which produce colored pixels by firing red, green, and blue electron guns at phosphors on the television or monitor screen.



Color Models - RGB

- Overlapping colors from 3 projectors produces new colors
  - Red + green -> yellow
  - Green + blue -> cyan
  - Red + blue -> magenta



Color Models - RGB

- **RGB color space** or **RGB color system**, constructs all the colors from the combination of the **R**ed, **G**reen and **B**lue colors.
- The red, green and blue use 8 bits each, which have integer values from 0 to 255. This makes 256*256*256=16777216 possible colors.

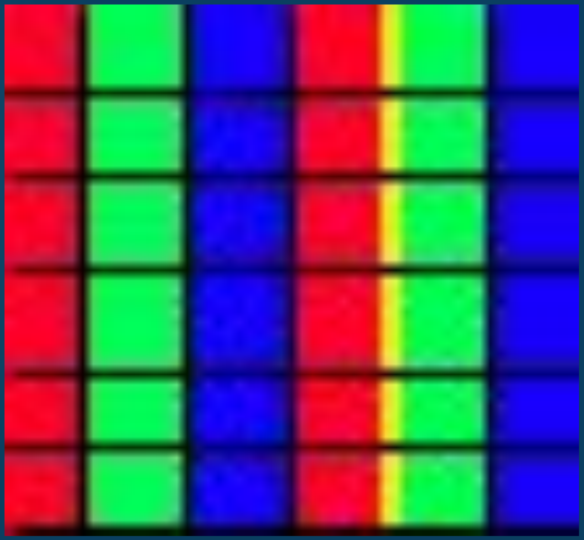
Color	HTML / CSS Name	Hex Code #RRGGBB	Decimal Code (R,G,B)
	Black	#000000	(0,0,0)
	White	#FFFFFF	(255,255,255)
	Red	#FF0000	(255,0,0)
	Lime	#00FF00	(0,255,0)
	Blue	#0000FF	(0,0,255)
	Yellow	#FFFF00	(255,255,0)
	Cyan / Aqua	#00FFFF	(0,255,255)
	Magenta / Fuchsia	#FF00FF	(255,0,255)

Color Models - RGB

- The color systems used by scientists and artists are entirely different.
  - An artist will mix blue and yellow paint to get a shade of green.
  - A scientist will mix green and red light to create yellow. The printed page in a magazine is yet another system.

Color Models - RGB

- This color model is used in computer monitors, television sets, and theater. If you put your eye up against your television screen you might something like

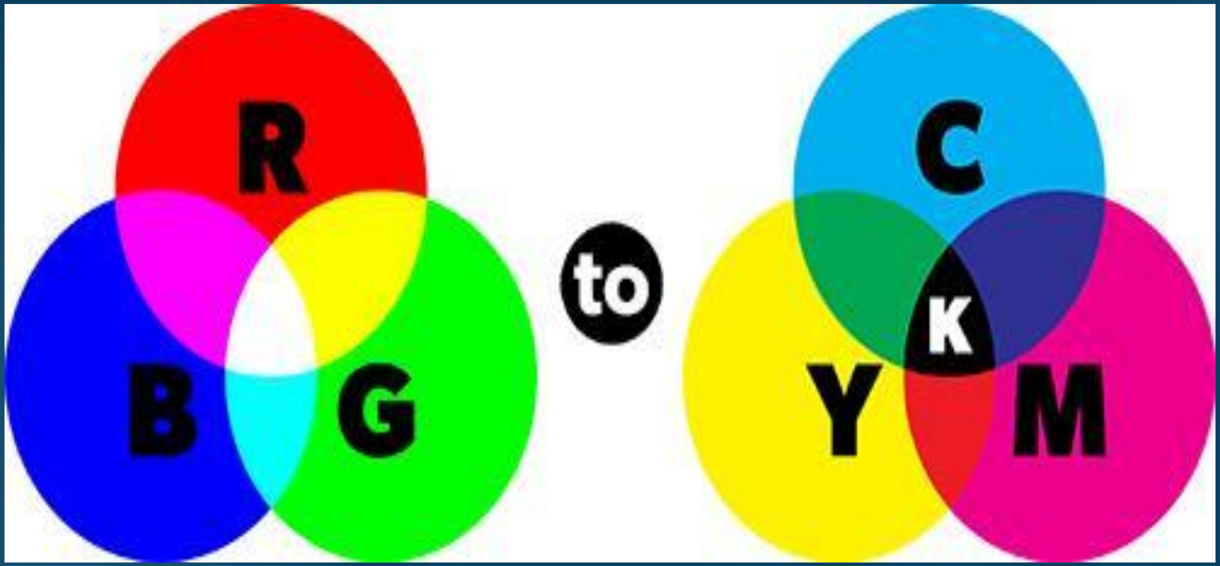


COMPLEMENTARY HUES – ADDITIVE MIXTURE

Complementary hues: Any hue will approach gray if additively mixed with it's opposite hue on the color circle. Such hue pairs are complementary. Of particular importance are the pairs that contain four unique hues: red-green, blue-yellow "complementary hues"

- Note that combining two complementary hues can never equal gray. For example, adding green to red will give you a yellowish gray, which is more *neutral* than the initial red

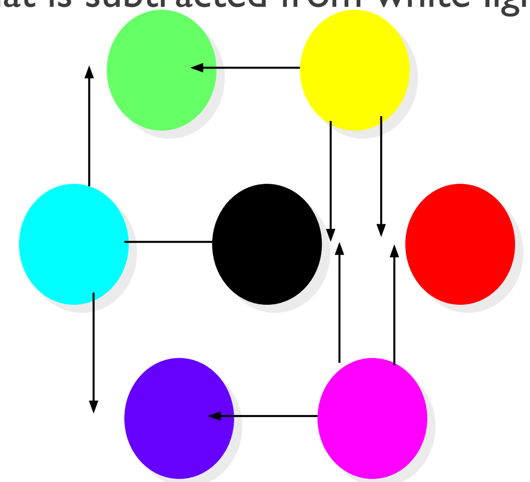
Color Models - CMYK



THE CMY(K) COLOR MODEL

- Used in electrostatic and in ink-jet plotters that deposit pigment on paper
- Cyan, magenta, and yellow are complements of red, green, and blue
- Subtractive primaries: colors are determined by what is subtracted from white light, rather than by what is added to blackness
  - white is at origin, black at (1, 1, 1):

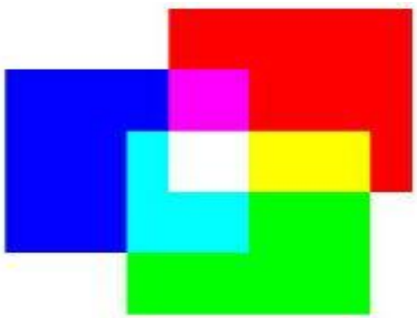
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



subtractive primaries (cyan, magenta, yellow) and their mixtures

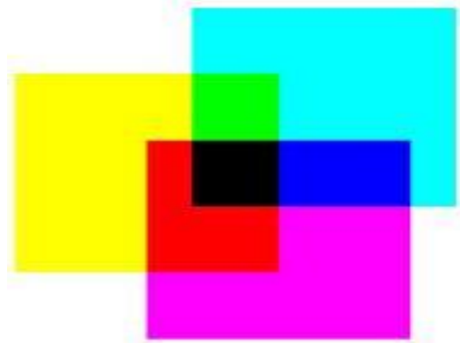
RGB AND CMYK

RGB and CMYK are two different color spaces. The RGB color space uses light in colors of red, green, and blue to create the visible spectrum. Our eyes see color in terms of reflected light, so the observed world is closer to RGB than CMYK. That is why native RGB devices that use light to create color, such as film recorders, scanners, and cameras can reproduce color fairly accurately.



RGB. Three colors of light, red, green and blue make white light. Cyan, magenta and yellow are also combinations of RGB.

The intensity of light also changes the color.



CMYK. Three inks, cyan, magenta and yellow make black. In practice this black lacks intensity, so a separate black (K) is usually added. Red, green and blue are made from CMYK.

CMY AND CMYK COLOR MODELS

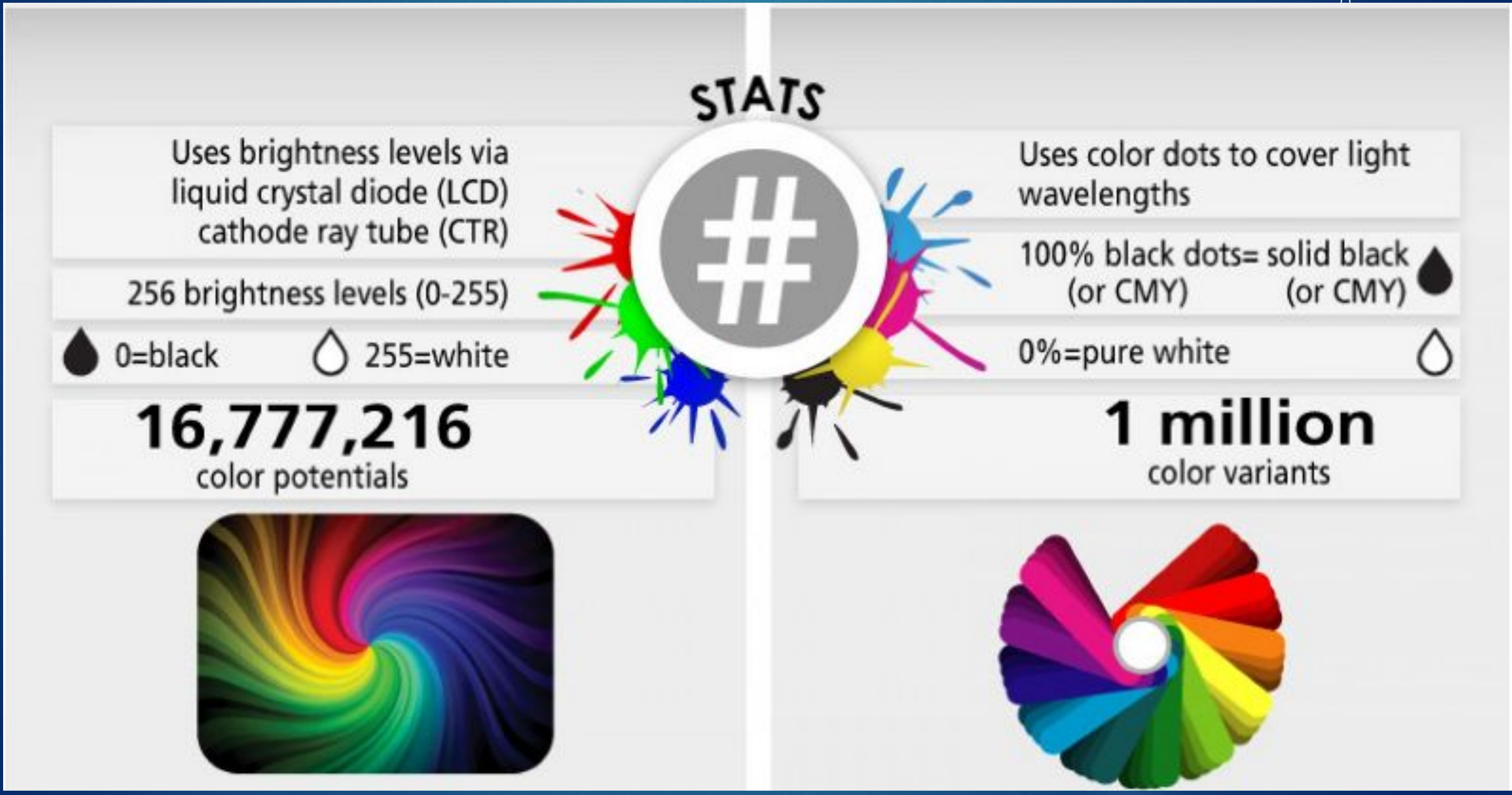
- **CMY:** Secondary colors of light, or primary colors of pigments are used.
- Used to generate hardcopy output (Printer and Copier).
- Some facts:
  - Printer papers are white (reflect all colors)
  - Printers use ink (Transparent)

- K (Blac**k**) is practical problem of C+M+Y≠Black (Muddy Brown) وذلك بسبب مشكلة في جودة الاحبار الكيميائية. Add a fraction of Black color

Color Models - CMYK

- Below is an example of a photo originally produced in RGB colors converted to CMYK colors as displayed on a computer monitor. Notice how the colors are much more vibrant on the RGB picture.

Color Models – RGB vs. CMYK



3-RGB AND HSV/HSI/HSL COLOR SPACE CONVERSION

- Human description of color is not RGB or CMY**K**

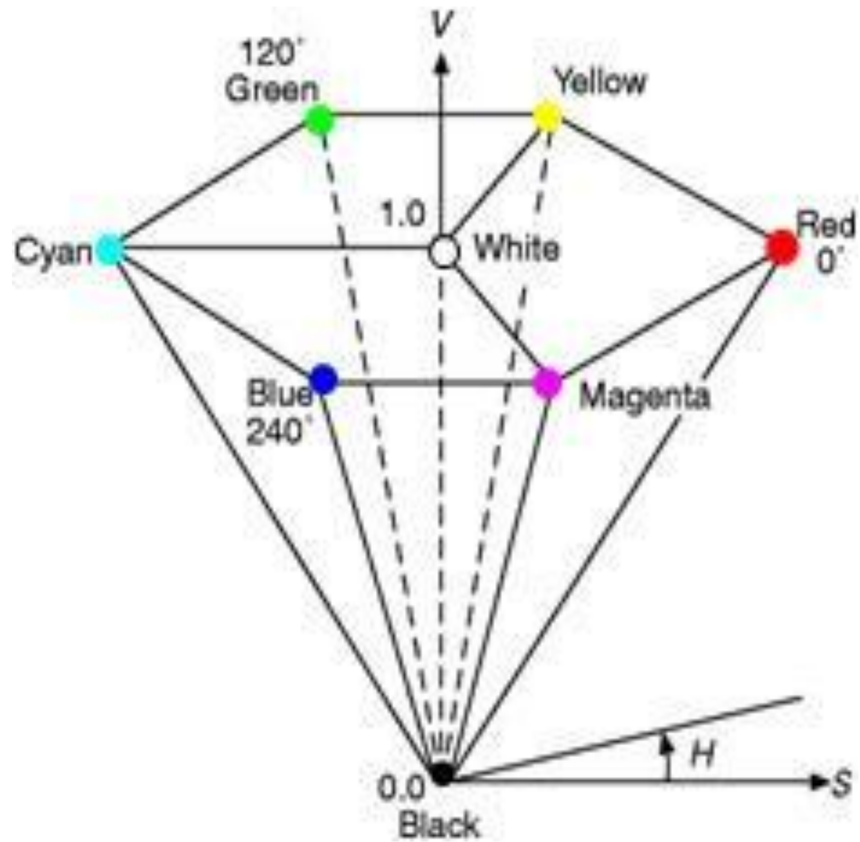
مثلاً: أحمر غامق، أخضر فاتح

- So, other models are required like: HSV/HSI/HSL
- HSV (hue-saturation-value), HSI (hue-saturation-intensity) and HSL (hue-saturation-lightness) are the three most common cylindrical-coordinate representations of points in an RGB color model.
- The HSV/HSI/HSL representations rearrange the geometry of RGB in an attempt to be more intuitive and perceptually relevant.
- The representations HSV, HSI and HSL are very similar, but not completely identical.

THE HSV COLOR MODEL- I

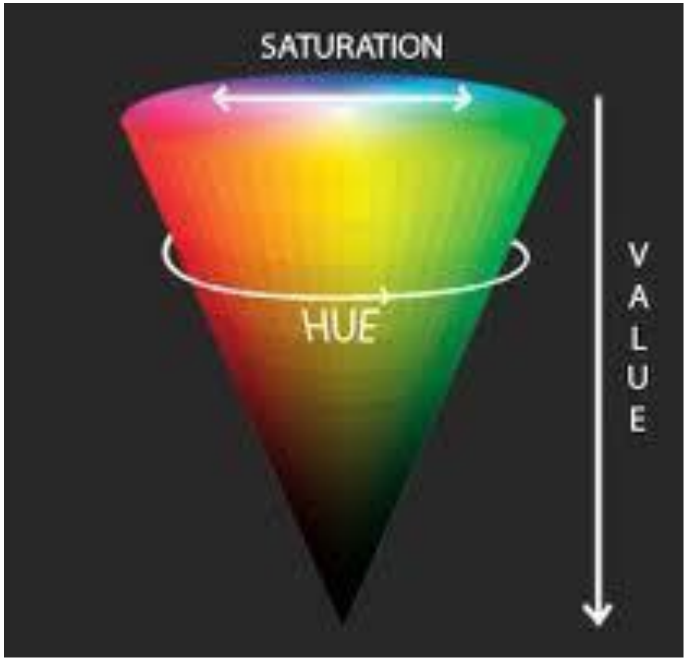
- **Hue**
- In HSV, hue represents color. In this model, hue is an angle from 0 degrees to 360 degrees.

Angle	Color
0-60	Red
60-120	Yellow
120-180	Green
180-240	Cyan
240-300	Blue
300-360	Magenta



THE HSV COLOR MODEL-2

- **Saturation**
- Saturation indicates the range of grey in the color space. It ranges from 0 to 100%. Sometimes the value is calculated from 0 to 1. When the value is '0,' the color is grey and when the value is '1,' the color is a primary color. A faded color is due to a lower saturation level, which means the color contains more grey.
- **Value**
- Value is the brightness of the color and varies with color saturation. It ranges from 0 to 100%. When the value is '0' the color space will be totally black. With the increase in the value, the color space brightness up and shows various colors.



https://www.tech-faq.com/hsv.html

Color Models

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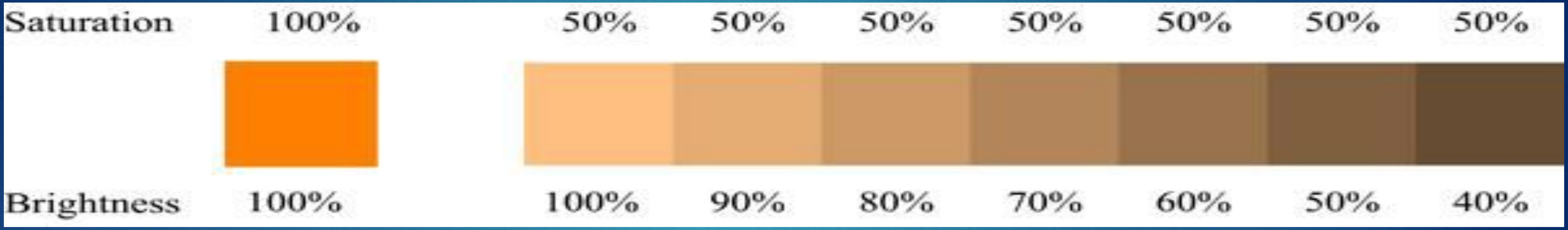
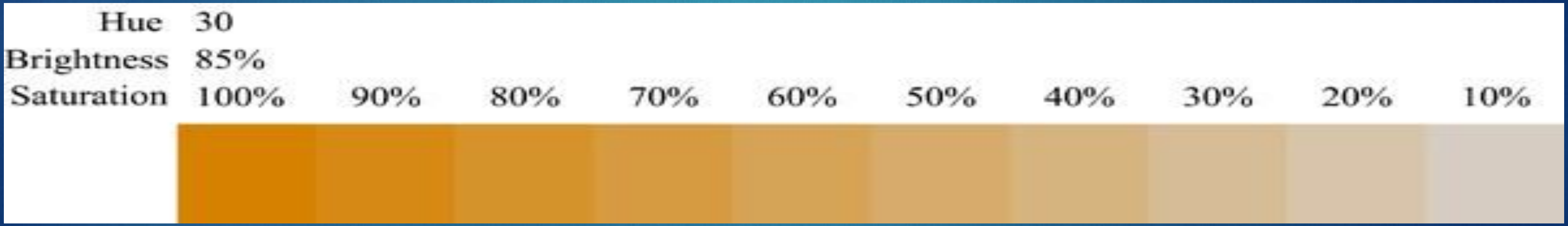
- Varying **Brightness**



- Varying **Saturation**



Suggested Rule for Creating **Color Palettes** using HSV

- Color **Palettes** is all selected colors to be used in scene.
- **Suggested rule (*not strict, you can ignore it sometimes*):** https://gamedevelopment.tutsplus.com/articles/picking-a-color-palette-for-your-games-artwork--gamedev-1174

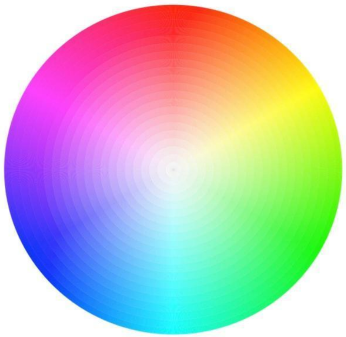
```
IF hues do not equal each other
THEN set saturations to match each other
AND set brightnesses to match each other
```

```
ELSE IF saturations do not equal each other
THEN set hues to match each other
AND set brightnesses to match each other
```

```
ELSE IF brightnesses do not equal each other
THEN set hues to equal each other
AND set saturations to equal each other
```

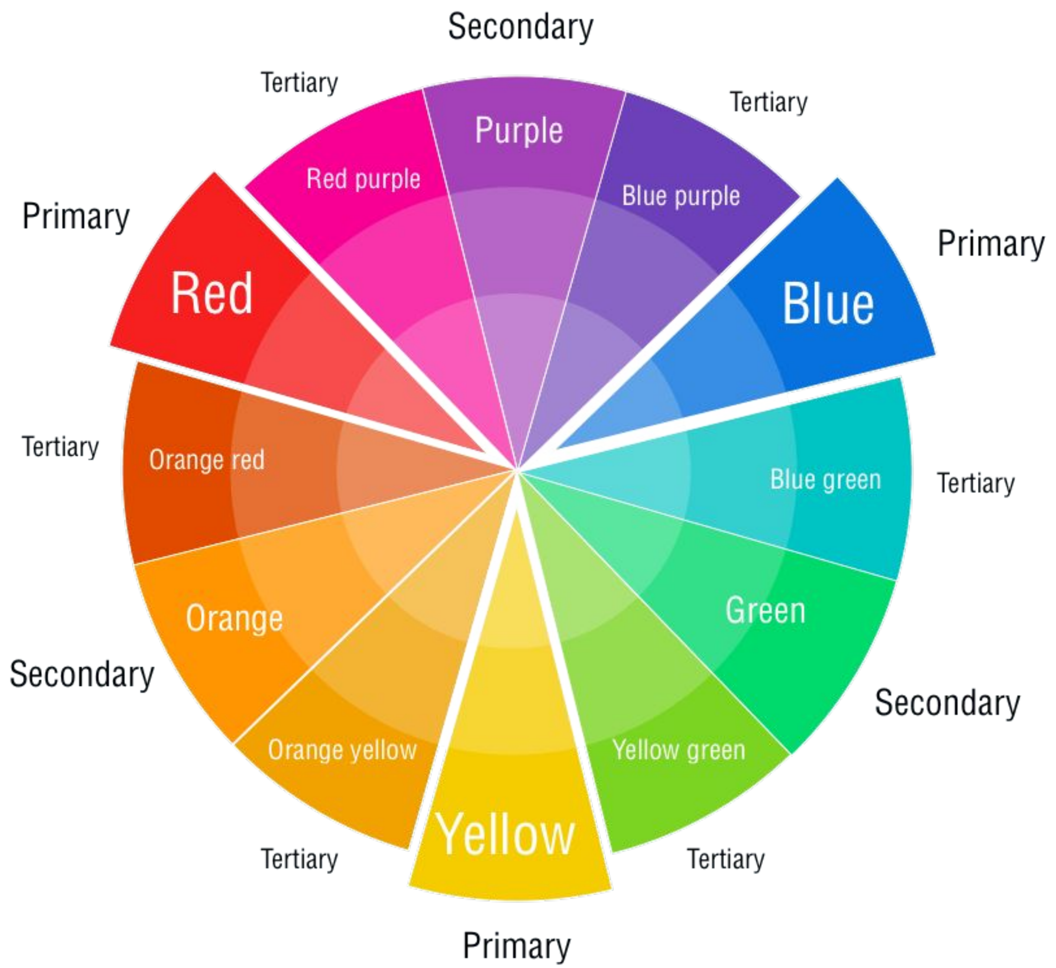
COLOR WHEEL

- An artist creates a color painting by mixing color pigmentsصبغة with white and black pigments to form the various shades in the scene.
-



You can use a color wheel to find color harmonies by using the rules of color combinations. Color combinations determine the relative positions of different colors in order to find colors that create a pleasing effect.

- Based on Color Wheel:



COLOR TEMPERATURE

- Even if you're a design beginner, you've likely heard the terms "warm, cool and neutral" terms. This is referred to as color temperature, and it's an essential consideration when it comes to color theory:
- **Warm colors** contain **shades** of **yellow** and **red**;
- **cool colors** **have a blue, green, or purple tint**;
- **Neutral colors** include **brown, gray, black, and white**.

- The temperature of a color has a significant impact on our emotional response to it. Within the psychology of colors, for example, warm colors show excitement, optimism, and creativity, whereas cool colors symbolize peace, calmness, and harmony.

Color Harmony Schemes

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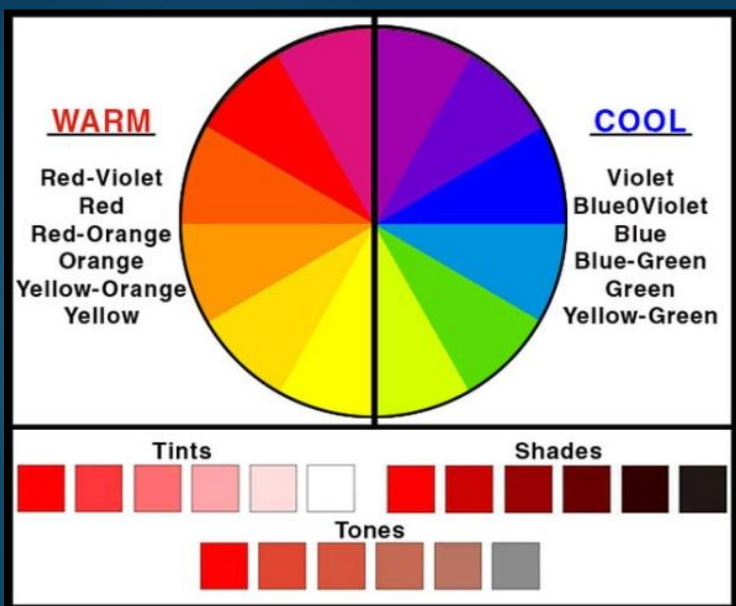
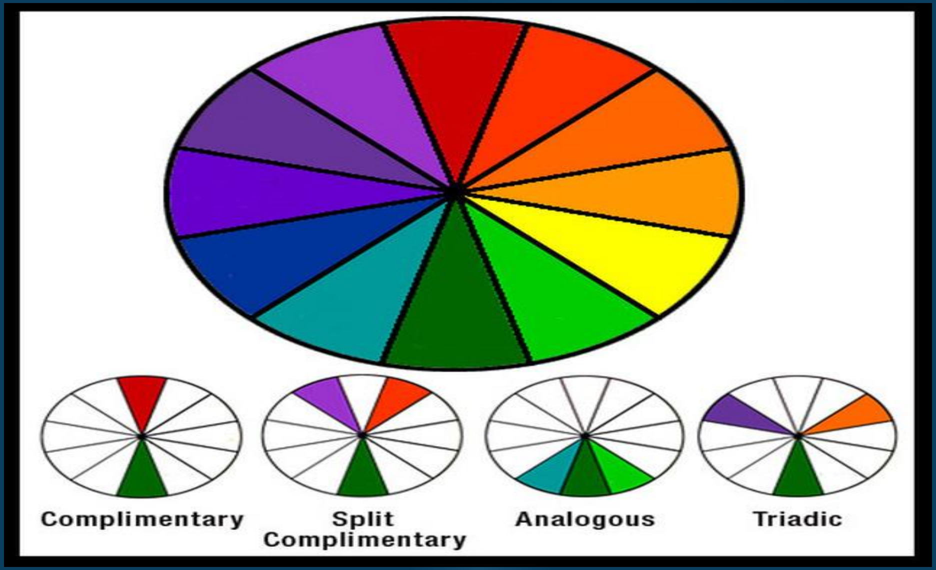
- Warm (yellow, orange, red) or cool (blue, green) colors



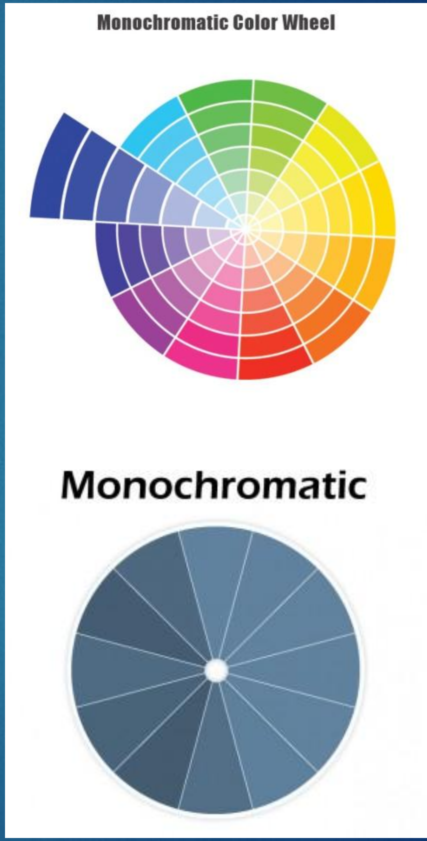
Color Harmony (Color Schemes)

- Certain combinations of colors tend to be pleasing. They arise from the color harmony schemes
- examples:
  - Monochromatic
  - Complementary
  - Analogous
  - Triadic

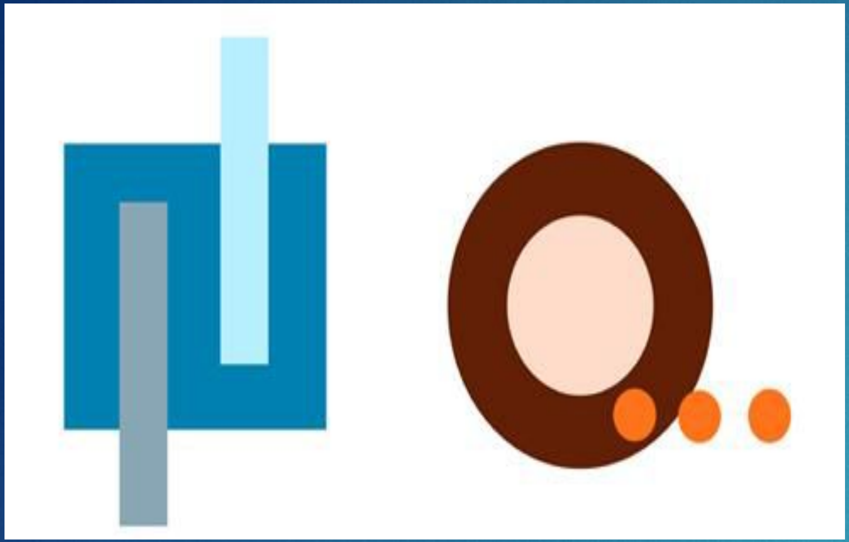


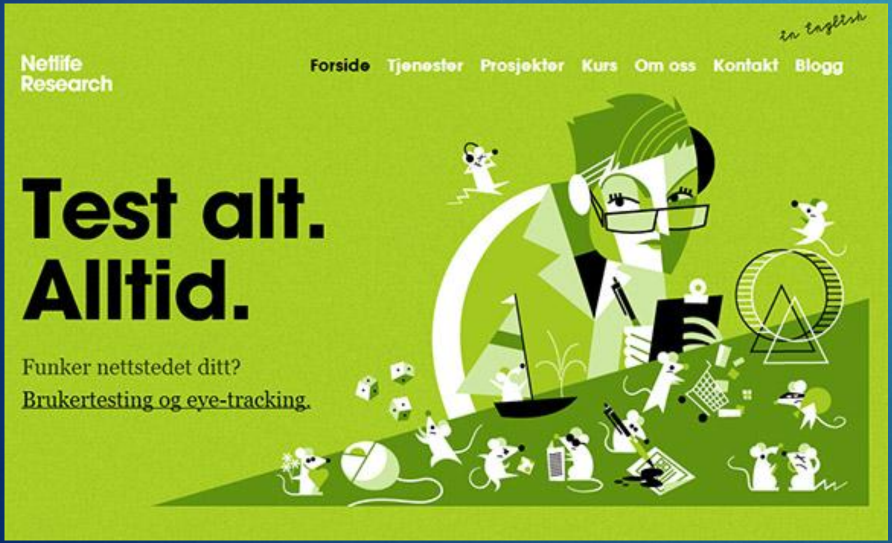
Harmony Schemes- 1-**Monochromatic**

- all colors have hues that are the same or within a few degrees of one another
- colors vary in saturation or brightness, but hue is consistent
- enhances cohesiveness to overall layout of web page



Monochromatic Examples

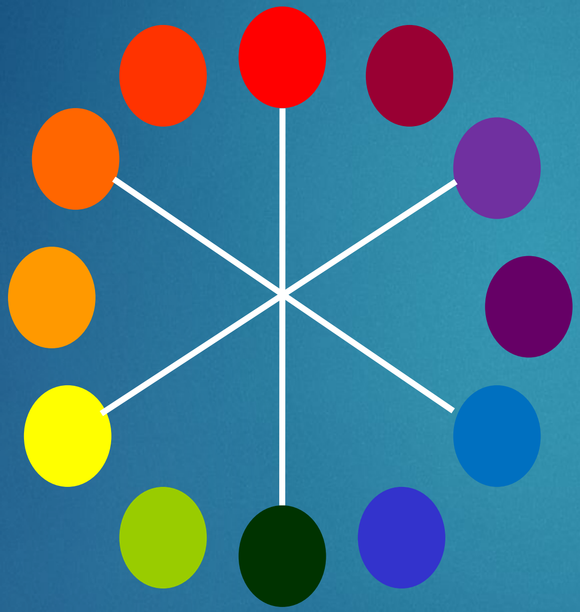




Harmony Schemes- 2-**Complementary**

- uses a pair of complementary hues, which appear opposite one another on a color wheel
- one color is dominant, the other is an accent
- use the dominant hue to fill the large areas

Using the Wheel

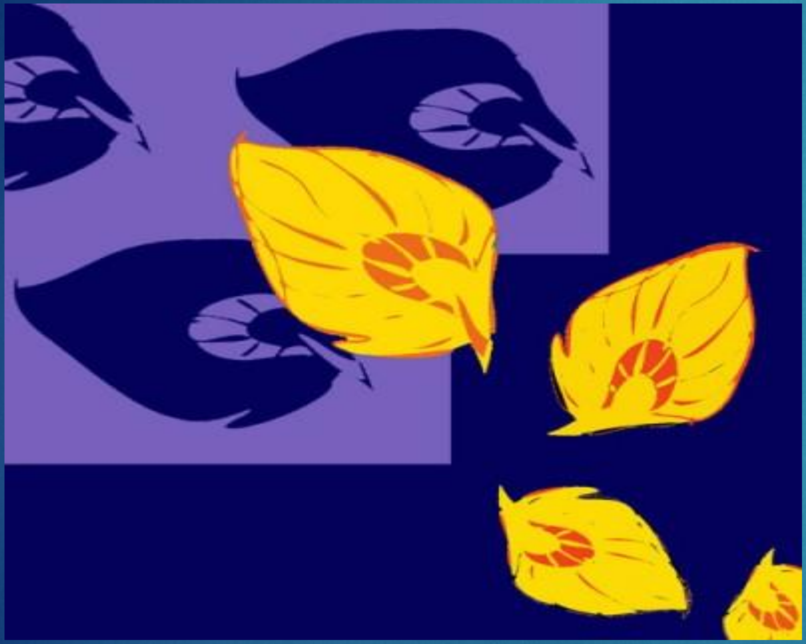


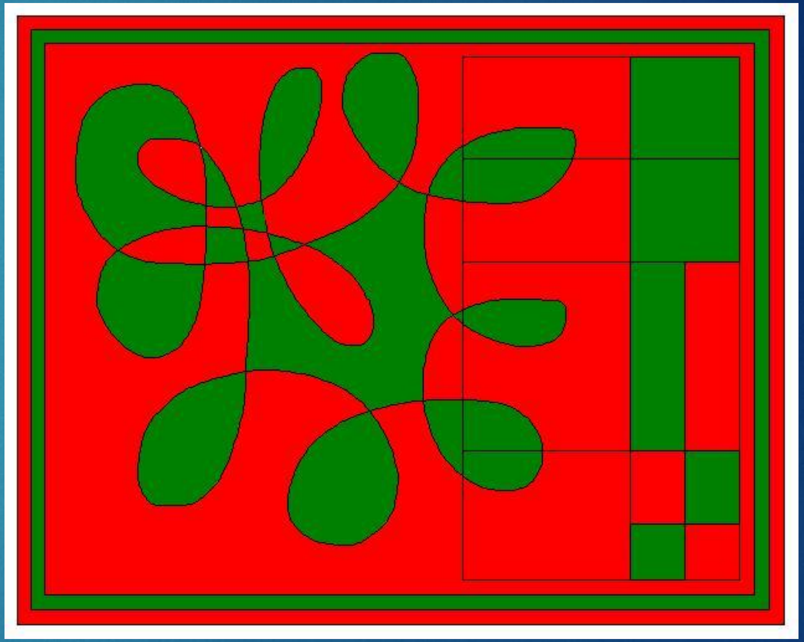
Complementary Colors are the colors opposite from one another on the wheel.

These colors provide the **most** visual contrast.

Contrast is the noticeable level of difference between two colors.

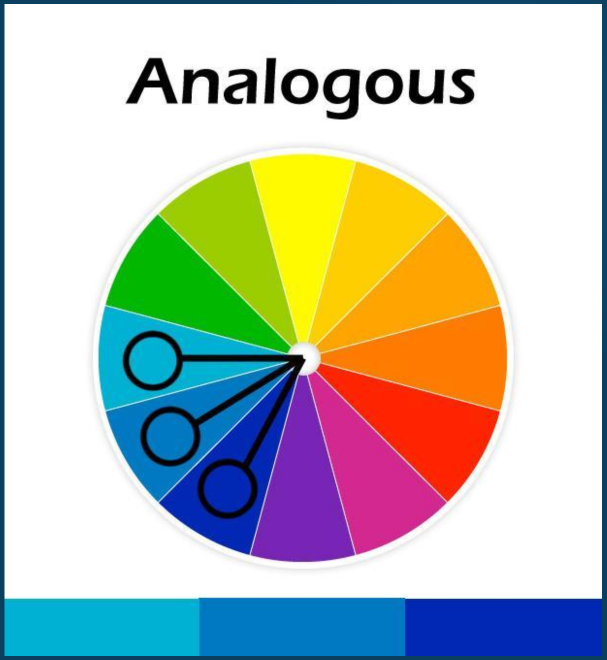
Complementary examples





Harmony Schemes- 3-**Analogous**

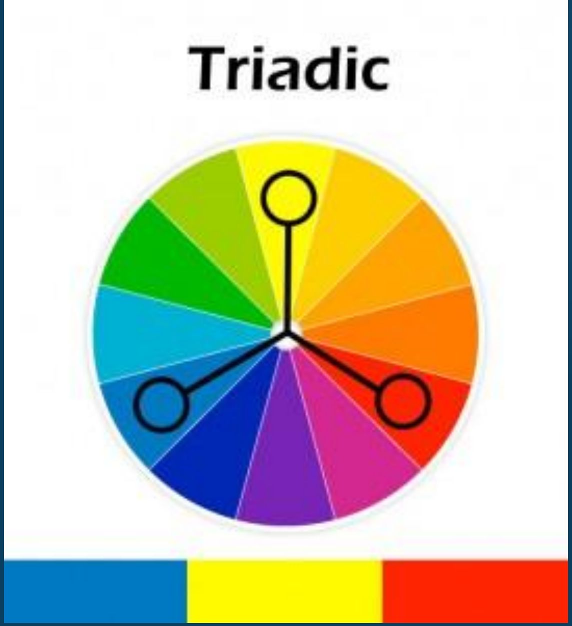
- three colors which lie close together on a color wheel
- often echo the colors found in nature
- pleasing combinations (such as orange, yellow, green)
- more interesting if the colors do not have the same brightness and saturation



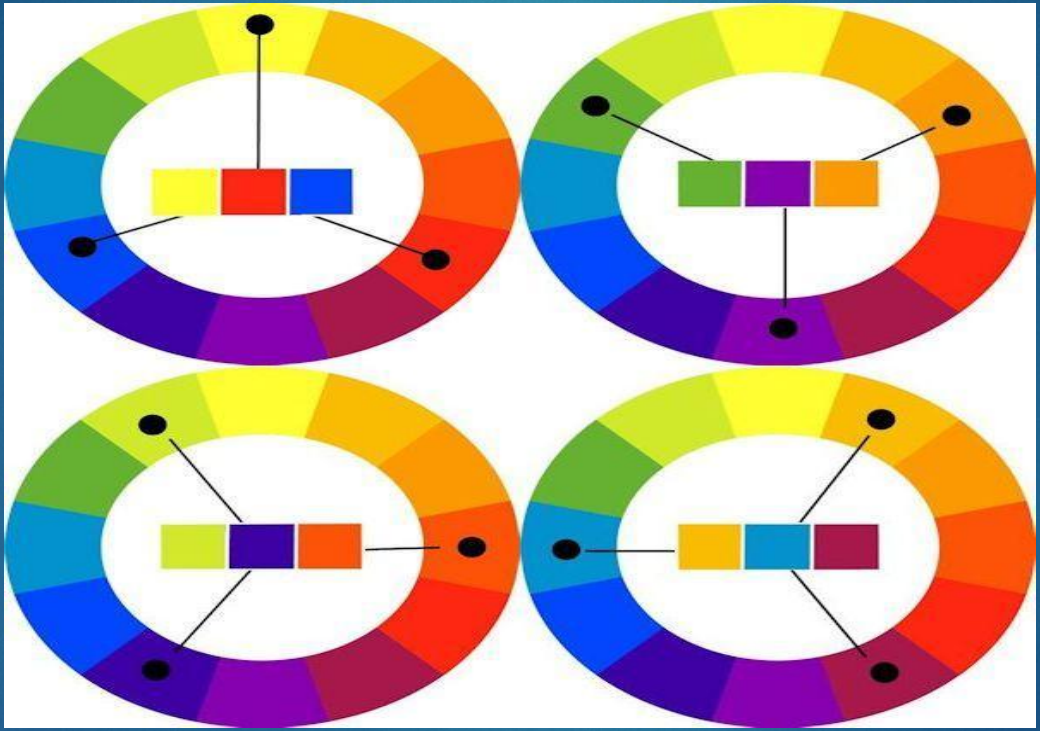
Analogous examples

Harmony Schemes- 4-**Triadic**

- Any 3 colors, spaced equally around a color wheel
- Color hues form a triad
- Offers wide variety of choice and can create excitement
- Can be overpowering unless colors chosen vary in brightness and saturation, or the number of text and background are limited



Triadic examples

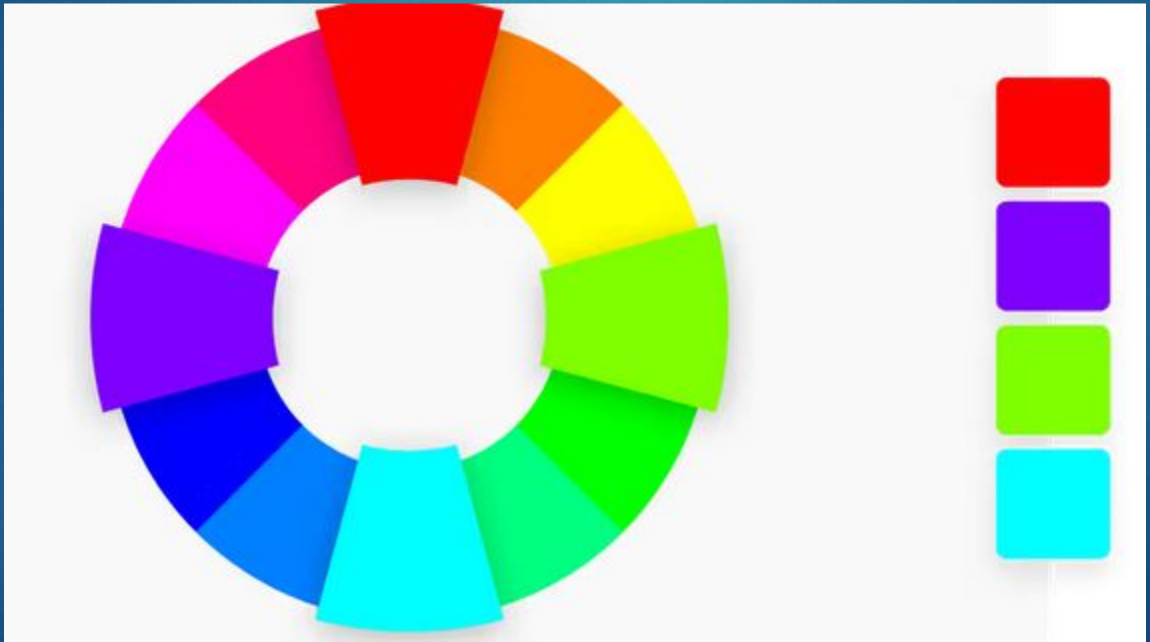


Triadic examples

Eyad Alsh

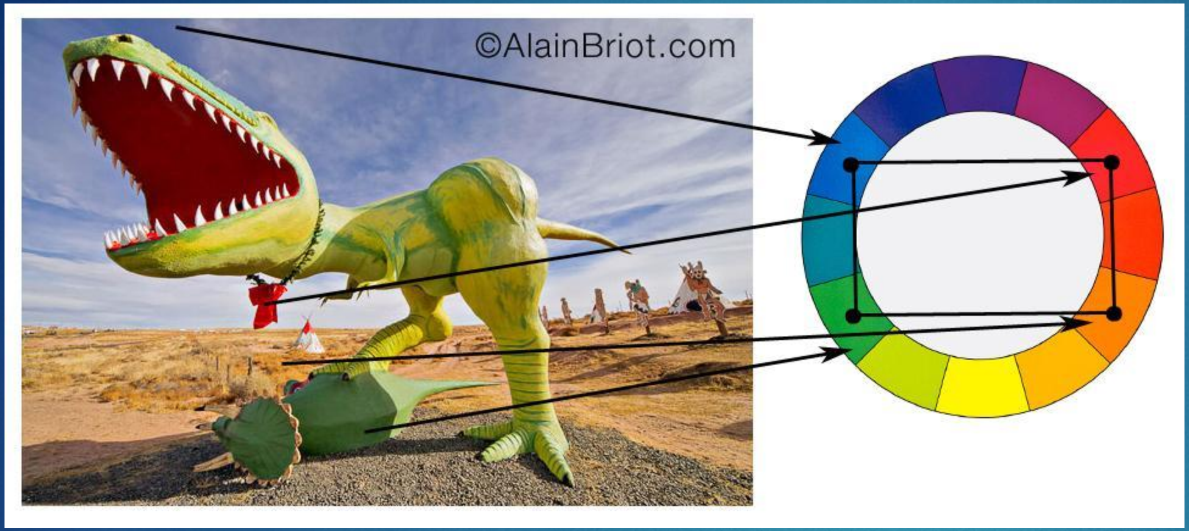
Harmony Schemes-
5- **Tetradic** رباعي

- Four colors that are evenly spaced on the color wheel. Tetradic color schemes are bold and work best if you let one color be dominant الغالب, and use the others as accents مساعدين . The more colors you have in your palette, the more difficult it is to balance,



https://www.canva.com/colors/color-wheel/

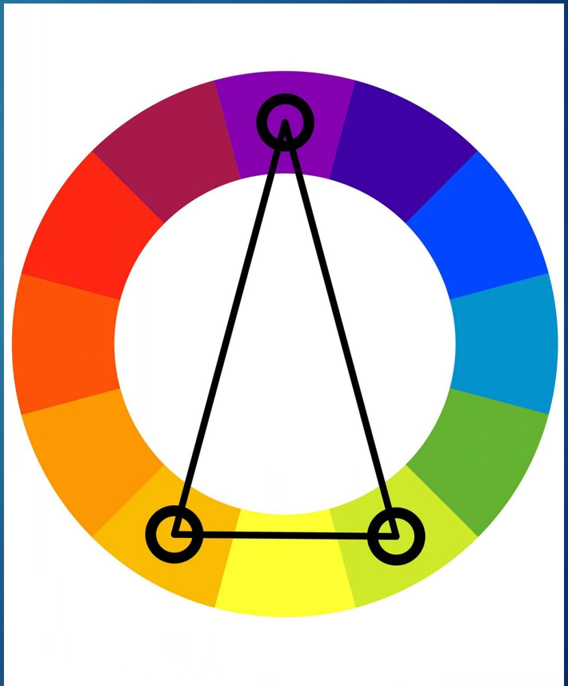
Tetradic رباعي examples



https://www.canva.com/colors/color-wheel/

Harmony Schemes-
6- **Split complementary**

- When figuring out split-complementary colors, you want to start out with a base color. From there, you combine it with two colors that sit directly adjacent to its complementary color without choosing the complementary color itself.



https://www.canva.com/colors/color-wheel/

Example for Split-Complementary

Eyad Alshareef

https://uxplanet.org/how-to-use-a-split-complementary-color-scheme-in-design-a6c3f1e22644

Color in Text and Background guidelines

1. Text should be readable مقروء
2. Contrast between text and background is important
3. Dark text on light background is best or one with high brightness and low saturation
4. Avoid combinations that differ only in their blue component (yellow on white)
5. Avoid red-green, red-blue, magenta-green combinations which cause vibration and eye fatigue.

Eyad Alshareef

Palette Flashing Problem
زغللة أثناء الانتقال بين صورتين

- Palette Flashing occurs when you use a series of images each with its own color palette. When the new image replaces the older one a flash occurs on the screen - a serious problem in multimedia
- Solution
  - use a single palette for all project images or
  - fade each image to white or black before showing the next image since white and black are present in most palettes

AN INTRODUCTION TO COLOR THEORY AND COLOR PALETTES

- **Primary colors**
  are colors you can’t create by combining two or more other colors. The primary colors are red, blue, and yellow.
- The **secondary colors**
  are orange, purple, and green—in other words, colors that can be created by mixing any **two of the three primary colors**.
- **Tertiary colors**
  are created by mixing a primary color with a secondary color. The tertiary colors are magenta, vermillion, violet, teal, amber, and chartreuse.

10 best color palette generators

- Coolors
- Adobe Color
- Paletton
- Colormind
- Color Hunt
- Canva
- Khroma
- ColorSpace
- Colorkuler
- Designinspiration

https://www.shopify.com/partners/blog/69878531-the-ultimate-list-of-online-colour-palette-generators-for-web-design

BIT DEPTH

- Image files are often referred to by the number of bits used to store color information
- A bit is a piece of binary data (meaning that it has only two possible values).
- It can be a one or a zero, true or false, yes or no
- It can be combined with other bits to produce a range of numbers

PIXELS AND IMAGES

No. of bits/pixel	No. of colors
1-bit-deep	only two colors (Black & White)
8-bit-deep	(2^8 =256) colors
24-bit-deep	(2^{24} =16777216) colors (true color)
36-bit-deep	(2^{36}) colors (High dynamic range -HDR)

BIT DEPTH

- One bit = two possible values (0, 1)
- Two bits = four values (00, 01, 10, 11)
- Three bits = eight values (000, 001, 010, 011, 100, 101, 110, 111)

- So, the number of possible values is 2^n
- where n is the number of bits per pixel

- Therefore, eight bits give 256 potential values
- Eight bits is one byte, so that is a common image format

BIT DEPTH

- Eight-bit images can be color or "grayscale"
- With grayscale, the values vary continuously from
- 0 (black) to 255 (white)
- With eight bit color, the colors are normally indexed.
- This means there is a look-up table containing RGB values for all the colors you want to show. This can be limiting, however.

COLOR DEPTH

- *Color depth*: the number of colors in an image.
  - The more colors the image uses, the more colors that the image needs to keep track of, and the more bits that are needed to store each pixel.

CHANGING COLOR DEPTH

- For a bitmap or raster image, changing the color depth changes the size of the file by a precise predictable amount, e.g.:
  - A 1 bit (black and white) image is one eighth the size of 8 bit
  - A 8 bit is one third the size of 24 bit
- With compression, these ratios can vary, but they're still a good rule of thumb

8 BIT COLOR
261X275 PIXELS, 38 KB GIF

4 BIT COLOR
261X275 PIXELS, 13 KB GIF

2 BIT COLOR
261X275 PIXELS, 5 KB GIF

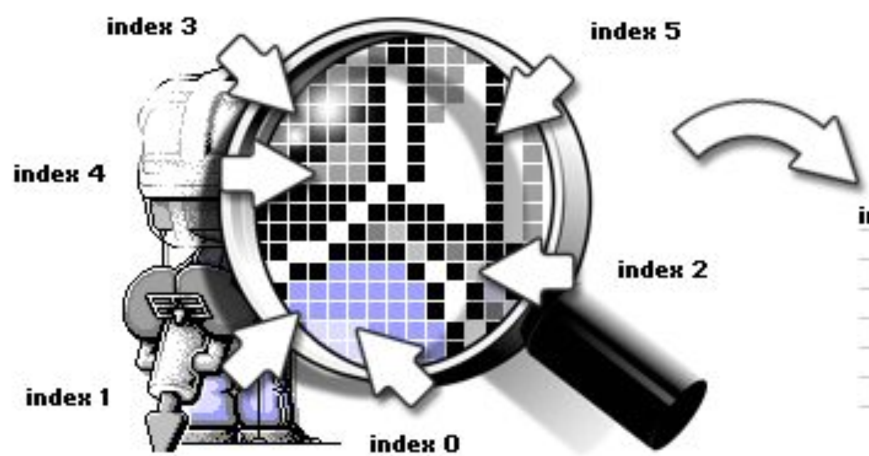
1 BIT COLOR
261X275 PIXELS, 2 KB GIF

HOW RASTER IMAGE STORE PIXELS COLORS?

- The are 2 techniques:
  - Indexed Color Images
  - True Color Image

INDEXED COLOR IMAGES

- *Indexed color images* use a so-called *lookup table* with a limited amount of colors.
- The maximum amount of colors in *GIF* images is, for example, 256.
- Every single *pixel* in a *GIF* image uses an index to designate its color. The index points to a specific color in the *lookup table*.
- By changing a color in the *lookup table*, all *pixels* with the same index will automatically change color.

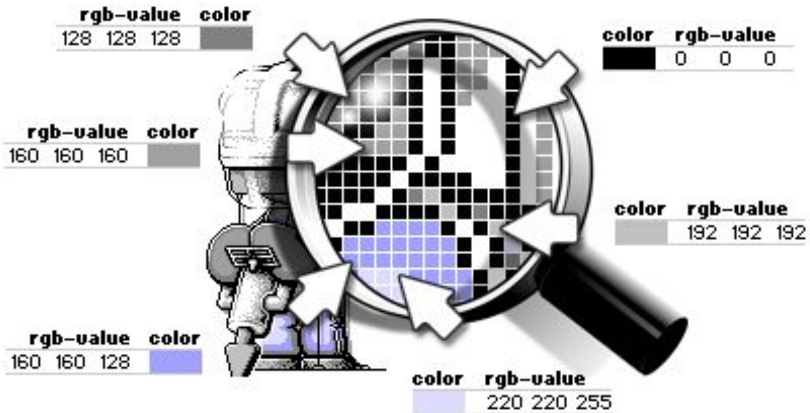


index	rgb-value			color
0	220	220	255	
1	160	160	128	
2	192	192	192	
3	128	128	128	
4	160	160	160	
5	0	0	0	

https://drububu.com/tutorial/image-types.html

TRUE COLOR IMAGES

- Compared to *indexed color images*, *true color images* lack a *color lookup table*. A *pixel* doesn't have an index referring to a specific color in the *color lookup table*. Every *pixel* has it's own (*RGB*) *color-value*, and, depending on the *file format*, a value for transparency (*RGBA*).



https://drububu.com/tutorial/image-types.html

IMAGE REPRESENTATION

- In the world of digital imaging, there are two types of graphical image files that you need to be aware of:
  - **Raster Image**
  - **Vector Image**

RASTER AND VECTOR FORMATS

There are 2 main graphic file formats:

Raster Formats (JPEG, BMP, TIFF, GIF, PNG, WebP, *PDF)

Vector Formats (AI, SVG, EPS, *PDF)

Note: There are new image formats being created every day for new and specific purposes.

*PDF’s can be either Raster OR Vector

Raster Formats

***Raster images are also called bitmap images** because they are made up of tiny squares called pixels; each 'pixel' contains exactly 1 color.*

Pros of Raster Formats

Rich Detail: Ever wondered what the term "dpi" stands for? It means "dots per inch," a measurement of how much detailed color information a raster image contains.

Say you've got a 1" x 1" square image at 300 dpi—that's 300 individual squares of color that provide precise shading and detail in your photograph. The more dpi your image contains, the more subtle details will be noticeable.

Pros of Raster Formats

Precise Editing: All of those individual pixels of color information can also be adjusted, one by one. So if you're some type of perfectionist, the level of editing and customization available in a raster image is almost unlimited.

Cons of Raster Formatsعيوبه

Blurry When Enlarged: The biggest downfall to raster images is that they become pixelated (aka grainy) when enlarged. Why is this? Well, there are a finite number of pixels in all raster images; when you enlarge a photo, the computer takes its best guess as to what specific colors should fill in the gaps. This interpolation of data causes the image to appear blurry since the computer has no way of knowing the exact shade of colors that should be inserted.

Cons of Raster Formats

Large File Size: Remember how a 1" x 1" square at 300 dpi will have 300 individual points of color information for the computer to remember? Well let's say you have an 20" x 20" photo— that's 120,000 bits of information for a computer to process which can quickly slow down even a fast computer.

JPEG

JPEG (Joint Photographic Experts Group) is a **lossy* compression method. Supports eight-bit grayscale images and 24-bit color images (eight bits each for red, green, and blue.) JPEG applies a lossy compression to images which can result in a <u>significant reduction of the file size</u>.

Keep in mind JPEG files suffer degradation when repeatedly edited and saved (it loses picture quality).

Recommended Use: When file size is more important than image quality. An example would be trying to save storage space on a website.

lossy* **compression is the class of data encoding methods that uses inexact approximations to represent content. Hence, there will be "loss" in exact quality.

BMP

The **BMP file format** (Windows bitmap) handles graphic files within the Microsoft Windows OS. Typically, BMP files are uncompressed, and therefore large and lossless; their advantage is their simple structure and wide acceptance in Windows programs.

Recommended Use: Large files (uncompressed), was used mostly for printing.

GIF

GIF (Graphics Interchange Format) is for common usage limited to an 8-bit palette, or 256 colors (while 24-bit color depth is possible).

The GIF format is **most effective for storing graphics with few colors, such as simple diagrams, shapes, logos and cartoon style images**, as it uses LZW lossless compression which is more effective when large areas have a single color, and less effective for images. Due to the GIF's format simplicity and age, it has almost universal software support. Because of its included animation capabilities, it is still widely used to provide image animation effects, despite its low compression ratio in comparison to modern video formats.

Recommended Use: GIF can make flawless copies at a high compression as long as the image contains large areas of uniform color, as long as the image has no more than 256 colors.

PNG

PNG (Portable Network Graphics) is a raster graphics **file** format that supports lossless data compression. **PNG** was created as an improved, non-patented replacement for **Graphics Interchange Format (GIF),** and is the most used lossless image compression format on the Internet.

Recommended Use: PNG was designed for transferring images on the Internet, not for professional-quality print graphics, and therefore does not support non-RGB color spaces such as CMYK.

Vector Formats

Vector images are the **No. 1 option when designing or creating a logo or illustration**. Because of the way images are created and saved, you will have more flexibility with making changes and be able to use your image at a variety of sizes.

You may only need a web logo now, but imagine how great it would be to have that image ready to use on a banner or merchandise later without having to create it all over again?

Pros of Vector Formats

Scalability: Vector data can be easily scaled and otherwise manipulated to accommodate the resolution of a spectrum of output devices. In simple terms: you can change the size and height/width ratio of the image without compromising the image quality.

Ease of Editing: Because every element in a vector file originated as a separate object it is very easy to edit!

Pros of Vector Formats

High Quality: The nature of vector files allows for almost zero pixilation when scaling the image.

Cons of Vector Formats

Hard Time Storing Complex Images: Vector files cannot easily be used to store extremely complex images, such as some photographs, where color information is paramount and may vary on a pixel-by-pixel basis.

Display Software: High-resolution raster displays are needed to display vector graphics as effectively.

Cons of Vector Formats

Expensive Software: Vector images can only be created using certain software: Adobe (Illustrator, InDesign, and Photoshop), Inkscape, DrawIT (Mac), Scribus, and Corel to name a few.

SVG

SVG (Scalable Vector Graphic) is the description of an image as an application of the Extensible Markup Language (XML). Any program such as a Web browser that recognizes XML can display the image using the information provided in the SVG format.

The SVG format enables the viewing of an image on a computer display of any size and resolution.

Recommended Use: Displaying images on screens (computers, lcd, smart phones, etc.)

PDF

PDF (Portable Document Format) in its simplest sense, is a file format used to present documents in a manner independent of application software, hardware, and operating systems.

A PDF can either be a Vector OR Raster file!

An easy way to see if your PDF file contains vector or bitmap content is to magnify the drawing to more than 800%. If you see smooth curves and straight lines, it's a vector file. If what you see on screen looks jagged, ragged or pixelated, it is a bitmap file (raster).

Recommended Use: When you want to present a file to someone that CAN'T be edited.

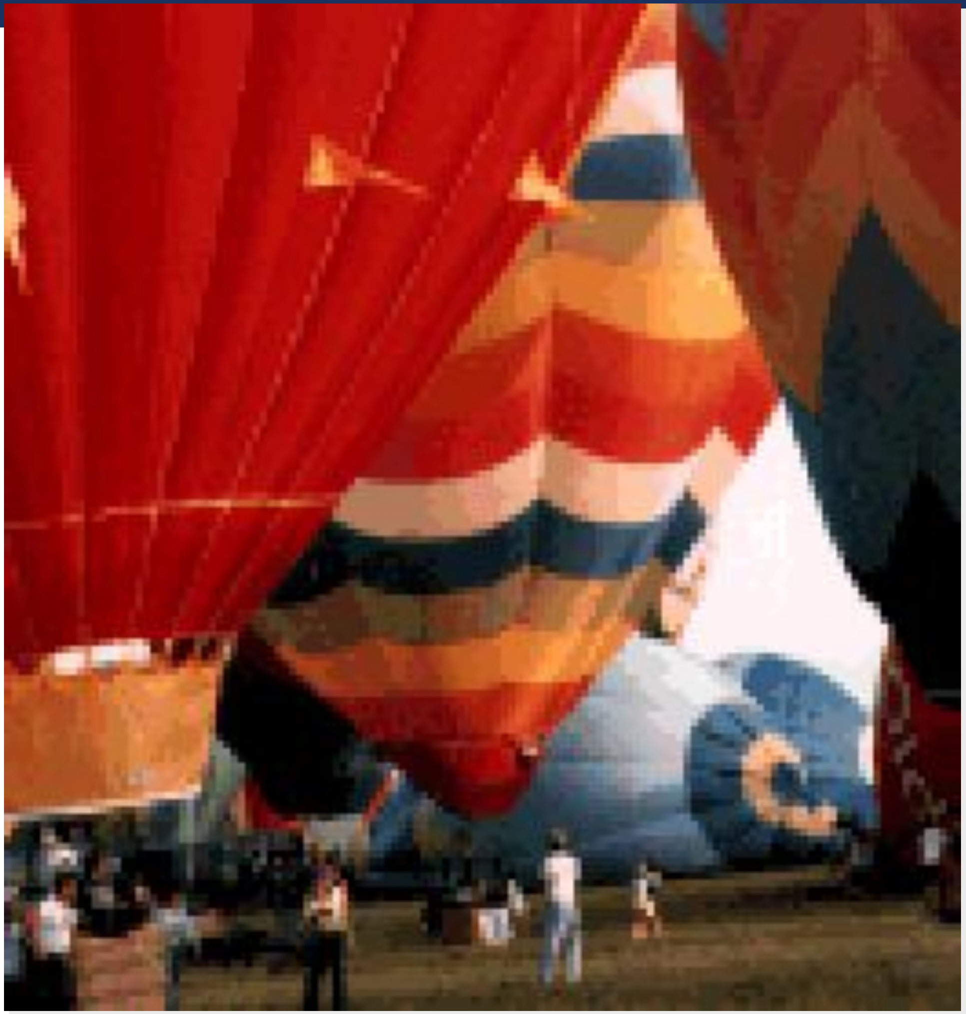
CHANGING RESOLUTION

- It is possible to greatly reduce the size of an image by reducing the number of pixels used to represent the image

- Size savings are squared; i.e. if you reduce an image from 200x200 (40,000 pixels) by half to 100x100 (10,000 pixels), the size savings is actually 1/2 * 1/2 = 1/4

8 BIT COLOR
261X275 PIXELS, 38 KB GIF

8 BIT COLOR
~130X137 PIXELS, 9KB GIF



8 BIT COLOR
~65X68 PIXELS, 3KB GIF

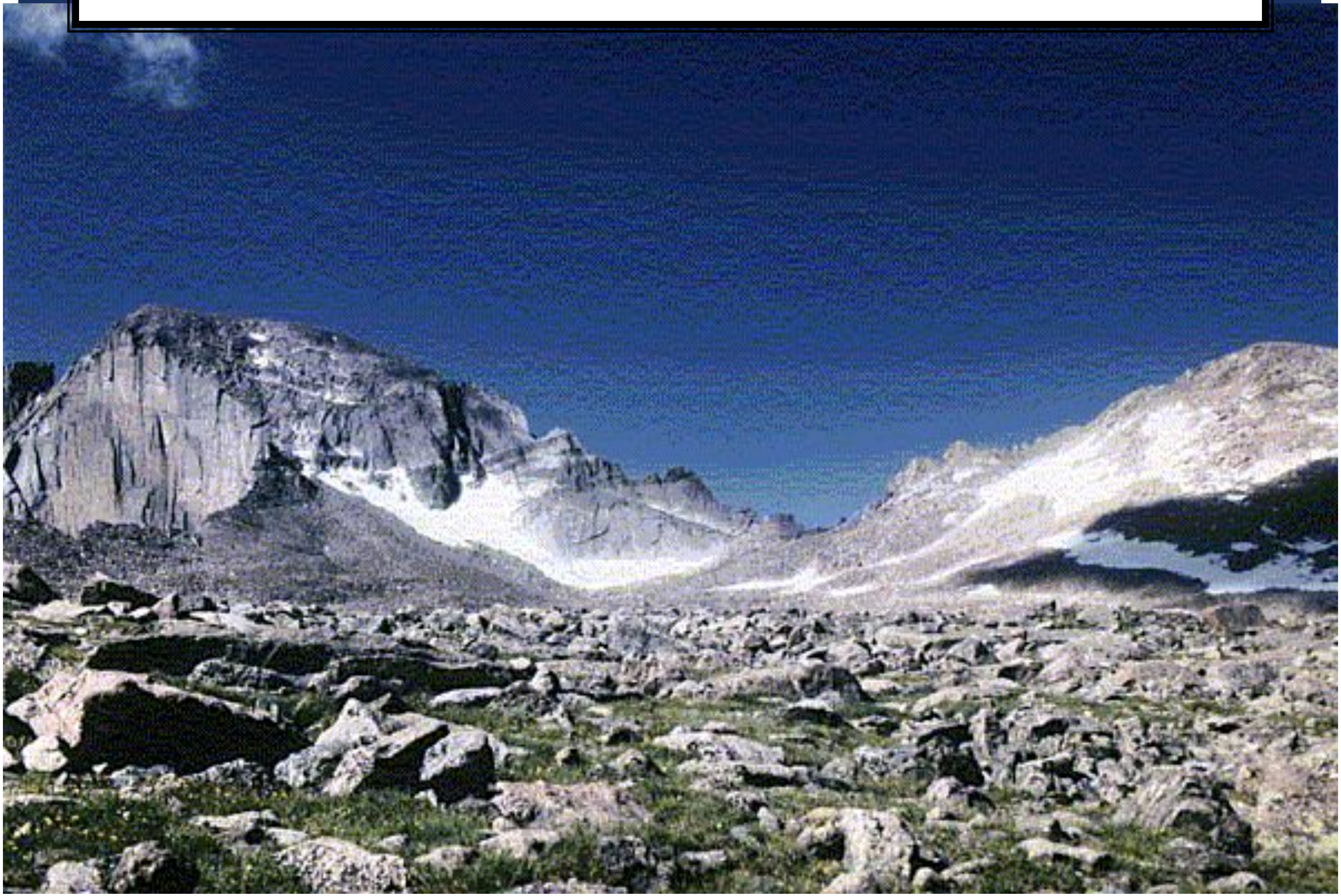
COMPRESSION

- Often, it is desirable to apply a compression algorithm to an image to reduce the file size

JPEG IMAGE, QUALITY 85; 59 KB

GIF IMAGE (8 BIT COLOR) 113 KB

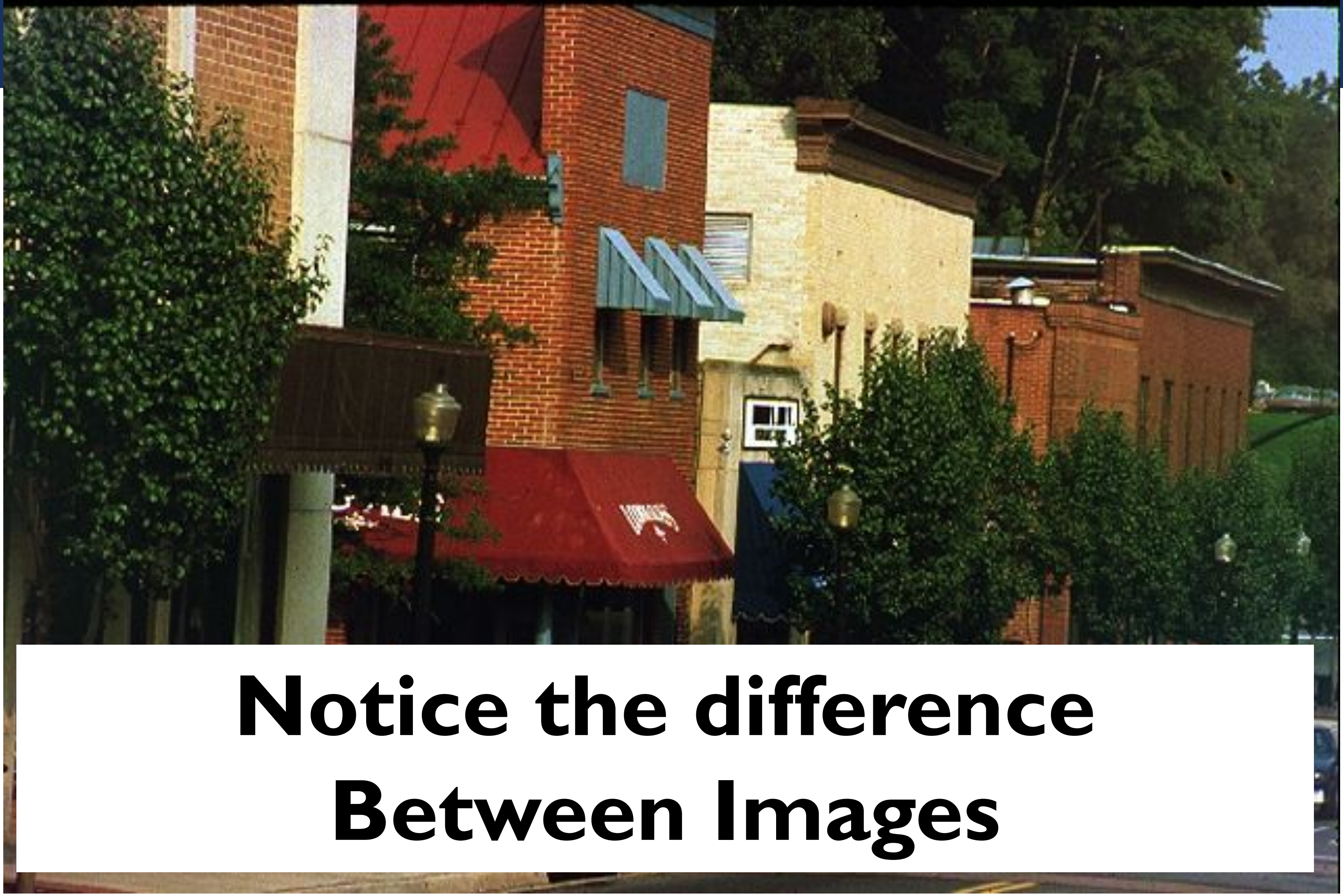
(DEPENDING ON ENCODING, PNG WOULD BE PERFECT BUT UP TO 6X AS BIG)



COMPRESSION EXAMPLES

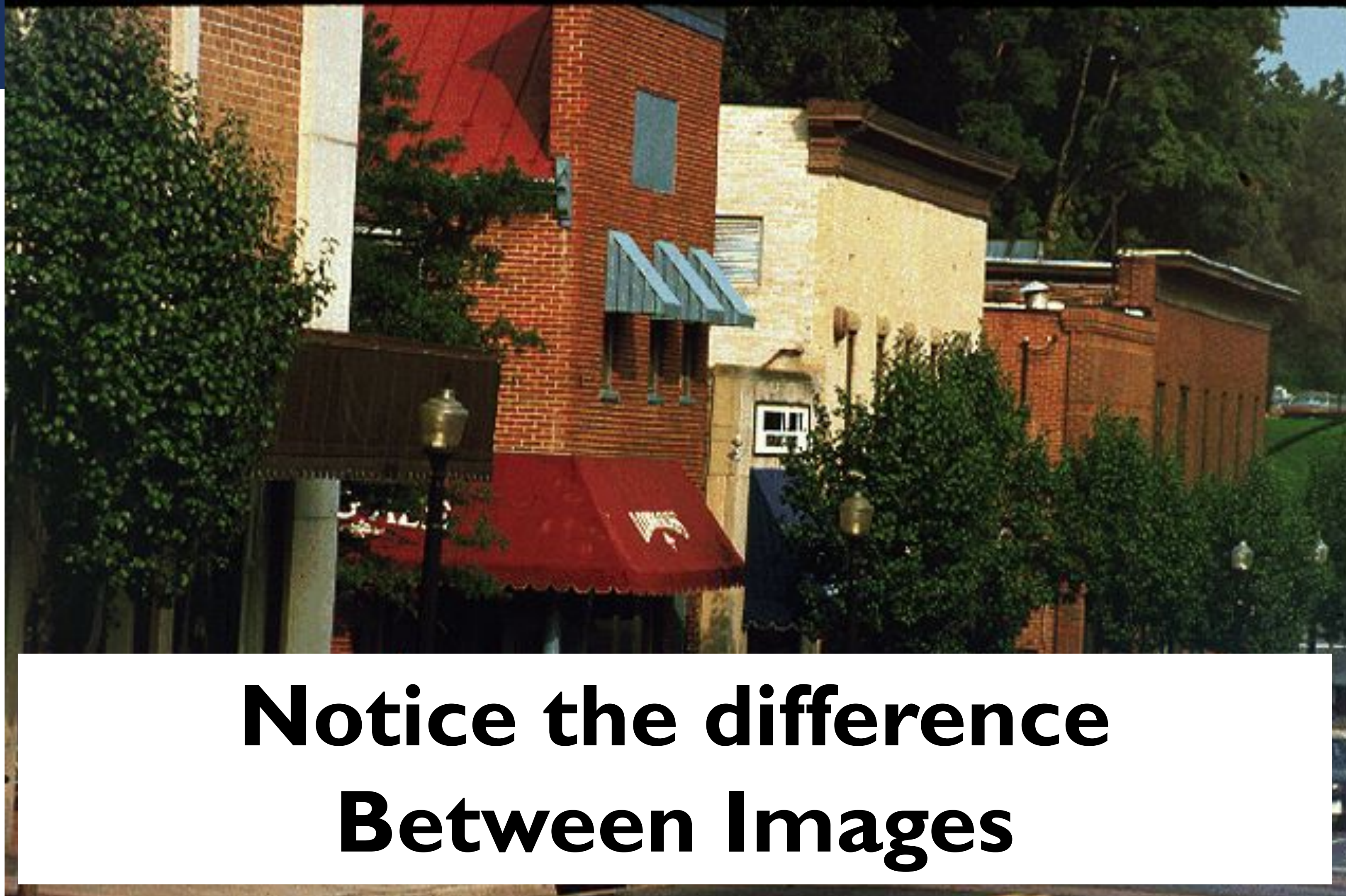
- Note :
  - **804 kb** **is a 24-bit bitmap image for the original image in the next example**

ORIGINAL (804 KB BITMAP)



Notice the difference
Between Images

GIF - 259 KB (8 BIT COLOR)

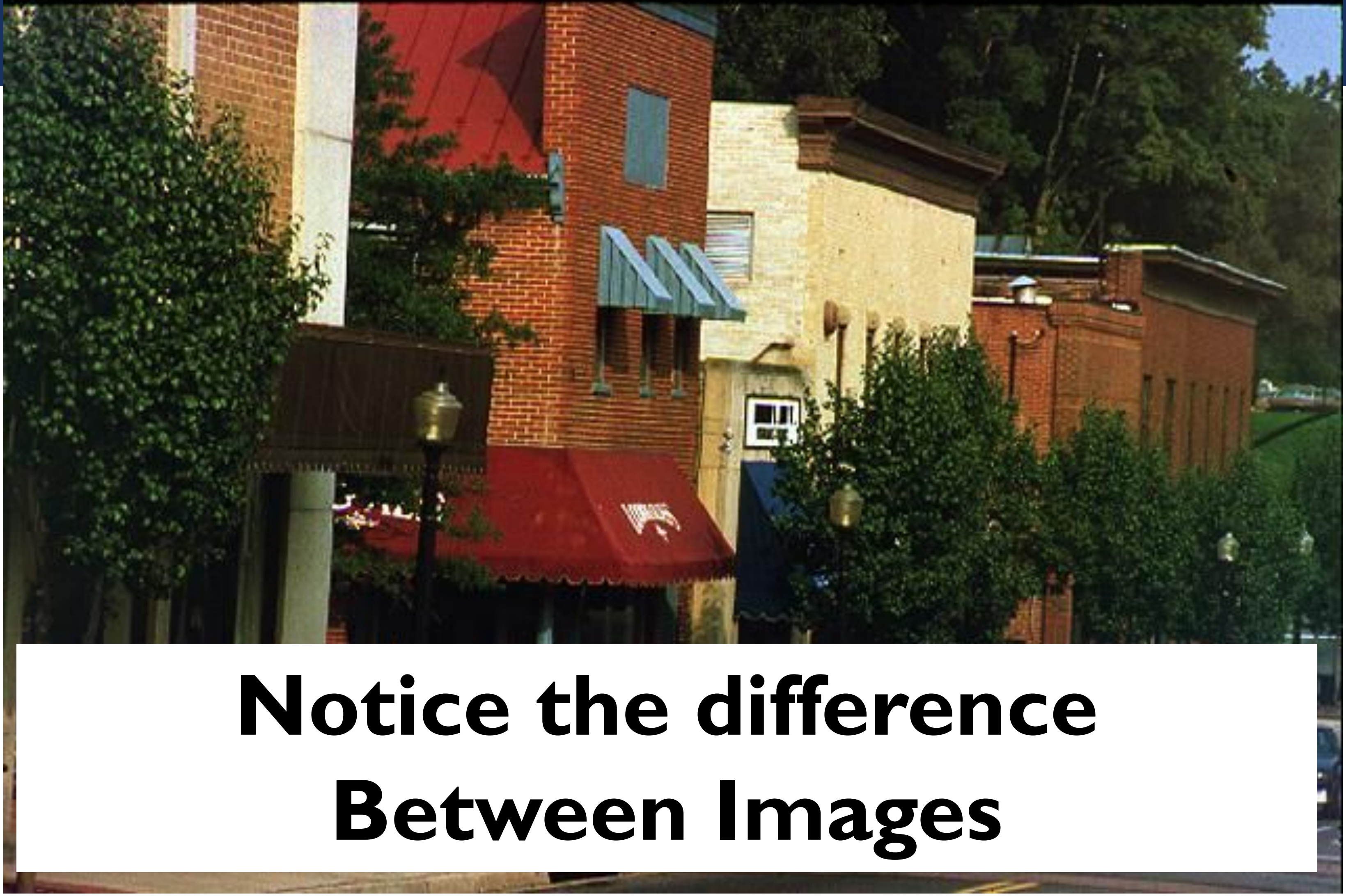


Notice the difference Between Images

JPEG - QUALITY 100 - 326 KB

Notice the difference
Between Images

JPEG - QUALITY 75 - 71 KB

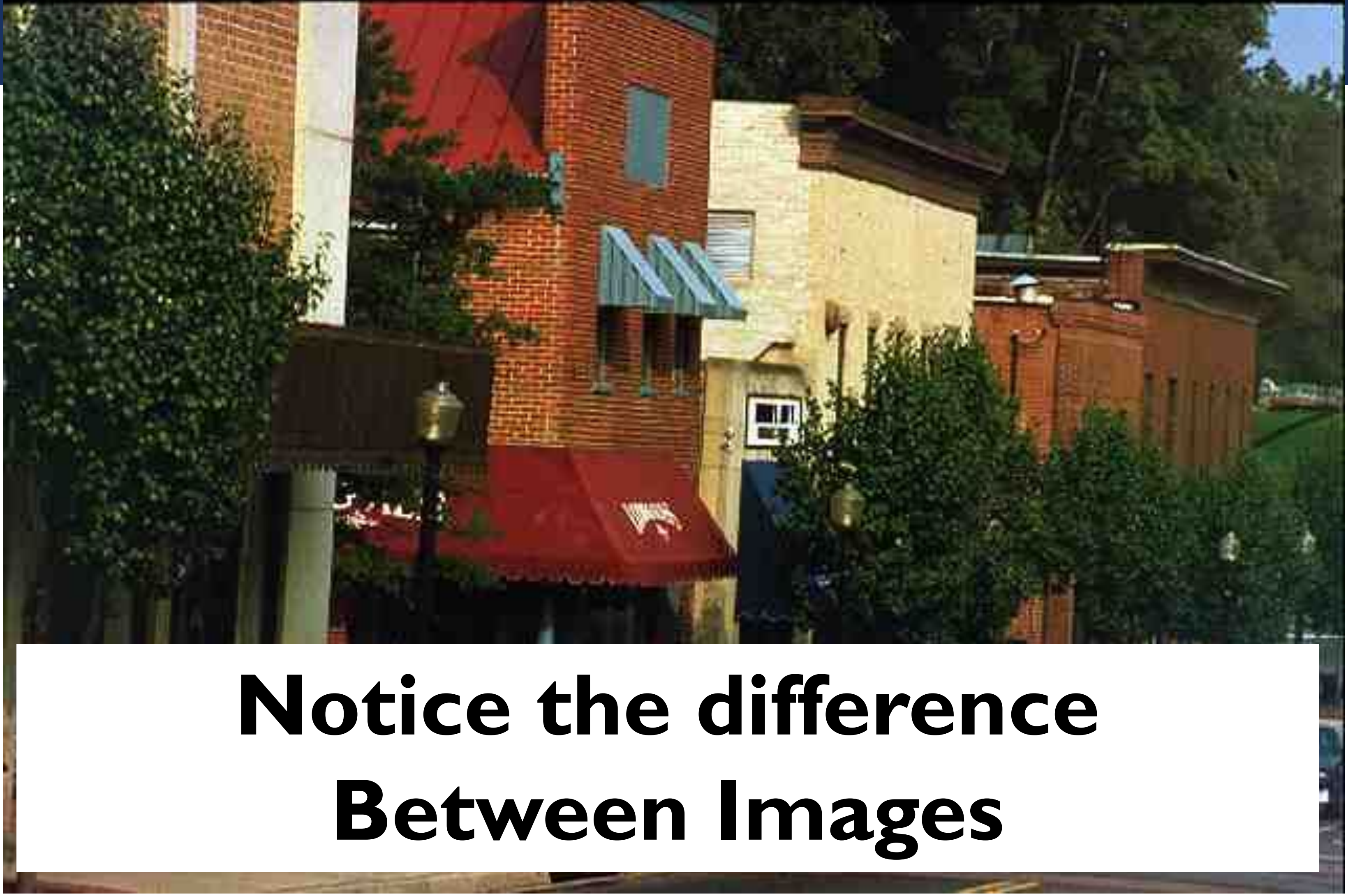


Notice the difference Between Images

JPEG - QUALITY 50 - 46 KB

Notice the difference Between Images

JPEG - QUALITY 25 - 29 KB



Notice the difference Between Images

JPEG - QUALITY 10 - 15 KB

Notice the difference Between Images

JPEG - QUALITY 5 - 9 KB

Notice the difference Between Images

TRANSPARENCY

- Some images include information on transparency encoded into the image.
- GIF images may include one transparent color in the color index table, allowing one “color” of the image to be transparent.
- PNG images can be indexed or true color and may include an additional byte of information per color (the alpha channel), which determines how transparent a particular pixel or color is.
- Native computer graphics formats are often 32-bit color – one byte each for RGB plus an additional byte for transparency or “alpha”

TRANSPARENCY

- ***Faked*** **"transparency":** the image is created and saved on a background to match the web page background.
  - Not always a perfect match on older computers or on jpegs.

TRANSPARENCY

- ***Real* transparency**: the image background is transparent, so the image floats on the web page background color.
  - Especially useful for patterned web page backgrounds.
  - Use the web page background color as the image background color while creating the image, but turn on transparency before saving the image to the web.

ASPECT RATIO

- *Aspect Ratio*: Ratio of width to height.
- *Never* change the aspect ratio of a photo – it distorts the image.

ASPECT RATIO

- Bad examples:

ASPECT RATIO

Resize, maintaining aspect ratio

- Good examples of resizing:

Crop to eliminate unnecessary background

CHANGING RESOLUTION

- It is possible to greatly reduce the size of an image by reducing the number of pixels used to represent the image

- Size savings are squared; i.e. if you reduce an image from 200x200 (40,000 pixels) by half to 100x100 (10,000 pixels), the size savings is actually 1/2 * 1/2 = 1/4

Editing Graphics

* Features for changing the sizes of images as well as their colors and other attributes. Some of these include:

Color correction

Cropping

Selecting

Scaling (resizing)

Layering

Filters

Cropping-

* When cropping an image, you are asking the software to show you only what you want to see and to get rid of other parts of the photo or image.
* In Photoshop, you would use it To draw a “box” around what you Want to keep.

Basics of image editing

Cropping an image

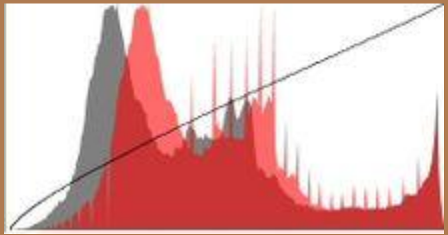
Digital editors are used to crop images. Cropping creates a new image by selecting a desired rectangular portion from the image being cropped. The unwanted part of the image is discarded. Image cropping does not reduce the resolution of the area cropped. Best results are obtained when the original image has a high resolution. A primary reason for cropping is to improve the image composition in the new image.

Basics of image editing

Histogram

Image editors have provisions to create an image histogram of the image being edited. The histogram plots the number of pixels in the image (vertical axis) with a particular brightness value (horizontal axis).
Algorithms in the digital editor allow the user to visually adjust the brightness value of each pixel and to dynamically display the results as adjustments are made. Improvements in picture brightness and contrast can thus be obtained

Sunflower image



Histogram of Sunflower image

Design & Developed by: EATL

Basics of image editing

Stamp Clone Tool

The Clone Stamp tool selects and samples an area of your picture and then uses these pixels to paint over any marks. The Clone Stamp tool acts like a brush so you can change the size, allowing cloning from just one pixel wide to hundreds. You can change the opacity to produce a subtle clone effect. Also, there is a choice between Clone align or Clone non-align the sample area. In Photoshop this tool is called Clone Stamp, but it may also be called a Rubber Stamp tool.

Original image

Image after stamp tool processed

Color Correction

* Photoshop has many color correcting features. Most of them can be found under the IMAGE>ADJUSTMENT menu. Examples include hue/saturation, levels, color input, Vibrance, brightness, contrast, etc.
* Red eye adjustment can be considered a color correction.

When color correction, a user is trying to “correct” a visual issue.

Selecting

- When you want to select certain images or parts of the image to edit or relocate.
- Photoshop features lots of selection tools- each for different situations.

(MAGIC WAND, QUICK SELECT, QUICKMASK, LASSO, RECTANGULAR & ELLIPTICAL MARQUEE, LAYER MASK, and PEN TOOL.)

Scaling

* Scaling is the basic resizing of an image.
* CONTROL>T allows the resizing of a specific object on a document. Holding SHIFT while tugging one of the four corners will keep your image from distorting.
* IMAGE>IMAGE SIZE allows you to resize that entire image as a whole.

Layering

* Layers are used in Photoshop to separate different elements of an image from others. This allows the user to treat them differently in many aspects.
* Since images are on a “different layer”, they will behave as separate entities but appear as one within the same composition.
* Changing the BLEND MODE will allow a certain amount of the layer directly underneath to come through. This can make a very interesting effect.

Filters

- Photoshop has a very extensive list of filters. They can be found under the FILTER>FILTER GALLERY menu. Additional filters and effects can also be found under the IMAGE>ADJUSTMENT menu as well.

Selection in Image Editing

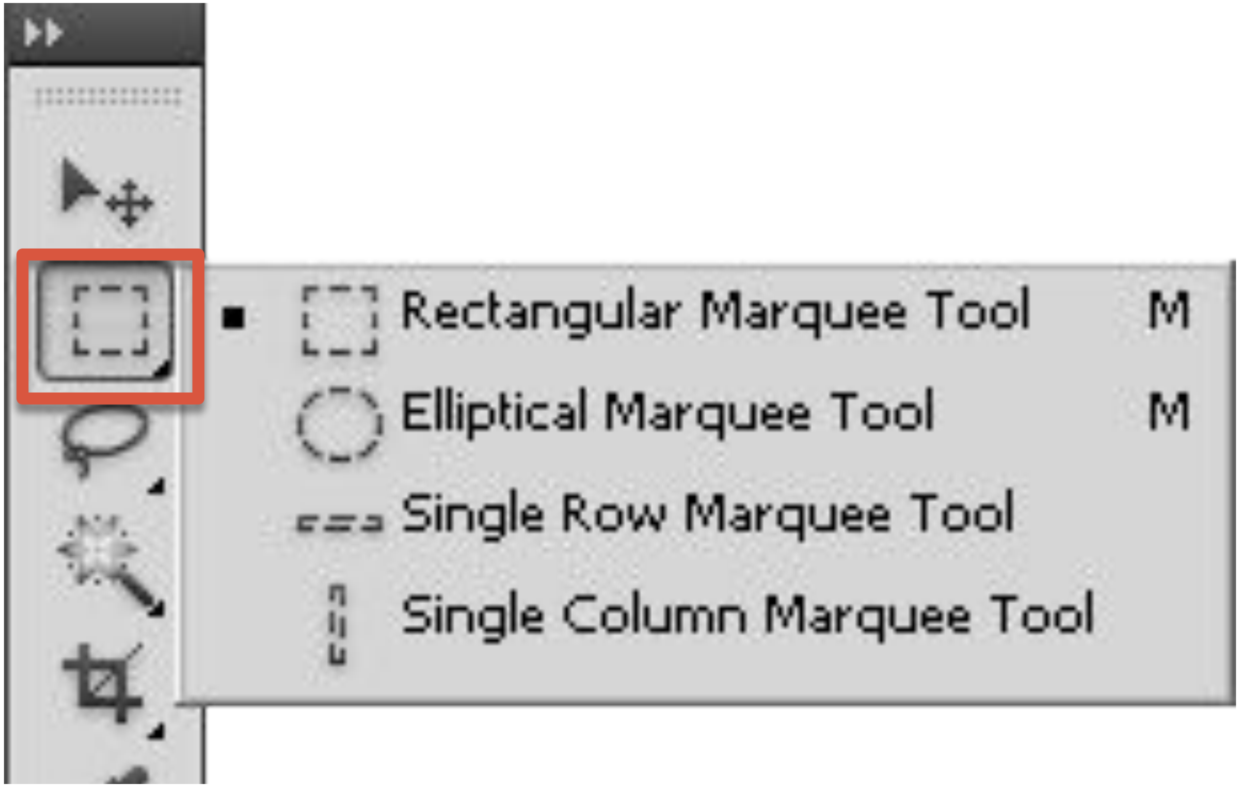
- Crucial in image editing
- Let you apply image effect (such as tonal or color changes) on the selected area
- Let you move the selected area
- The nonselected area is protected from the alteration

Categories of Selection Tools in terms of the way they are designed to work

* Predefined shapes
* Lasso
* By color (http://www.youtube.com/watch?v=6kHR419NjLQ)
* By painting with a brush (http://www.youtube.com/watch?v=zqYkTyG1PfA)
* By drawing an outline around the area

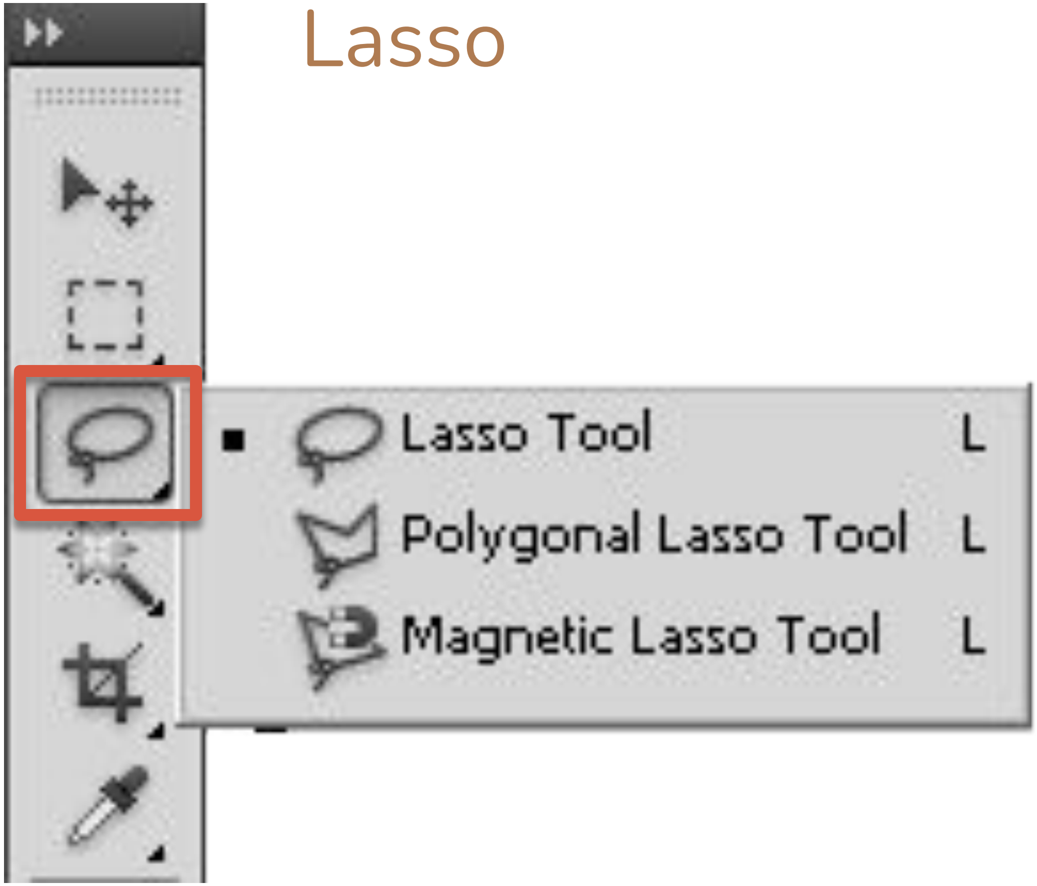
Predefined Shapes

Marquee tools



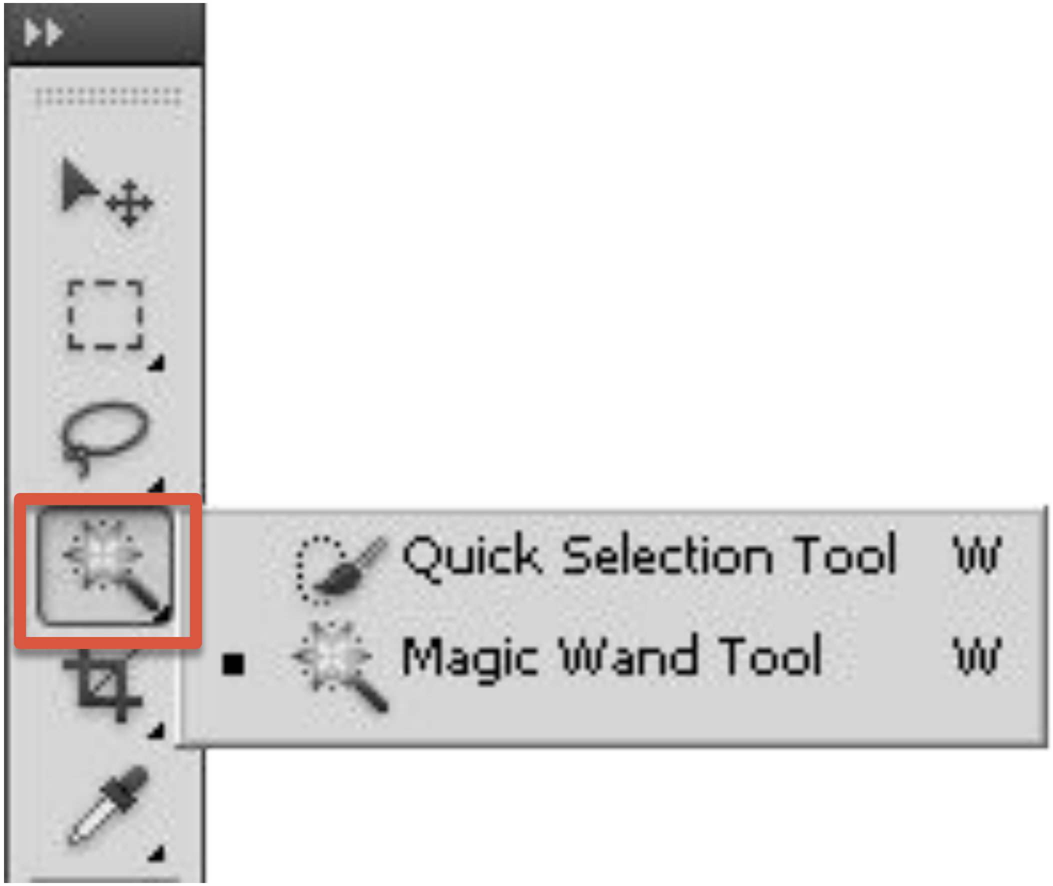
Lasso

Lasso tools

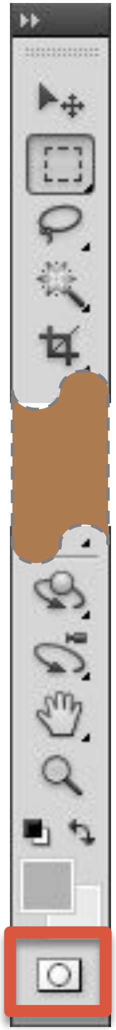


By Color: Magic Wand

Magic Wand



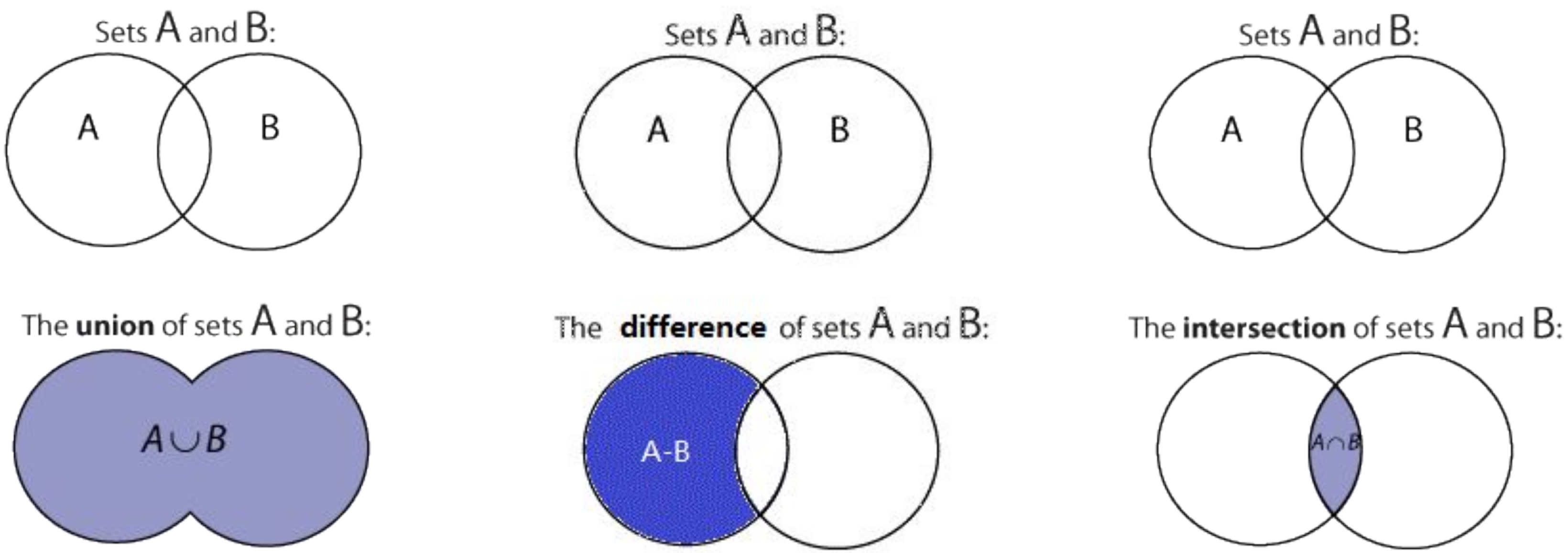
By Painting with a Brush



Edit in Quick Mask Mode

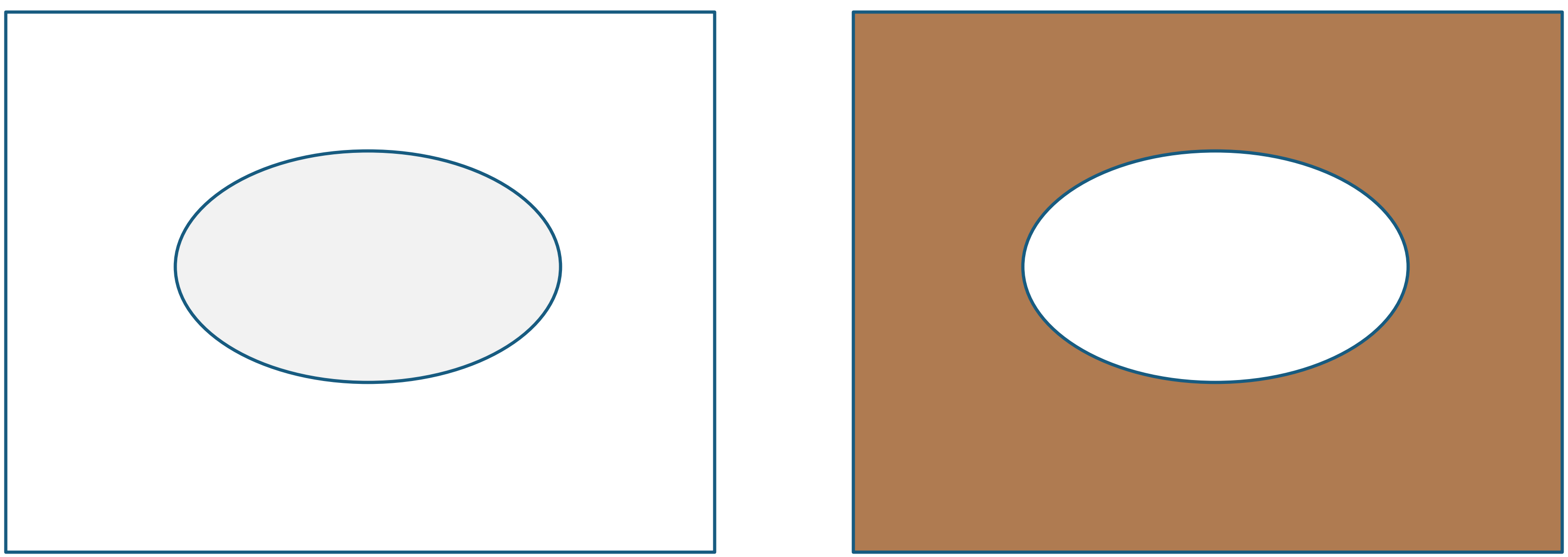
Combining Selections

* Add (Union)
* Subtract (Difference)
* Intersect



Inverting Selection

* Select > Inverse



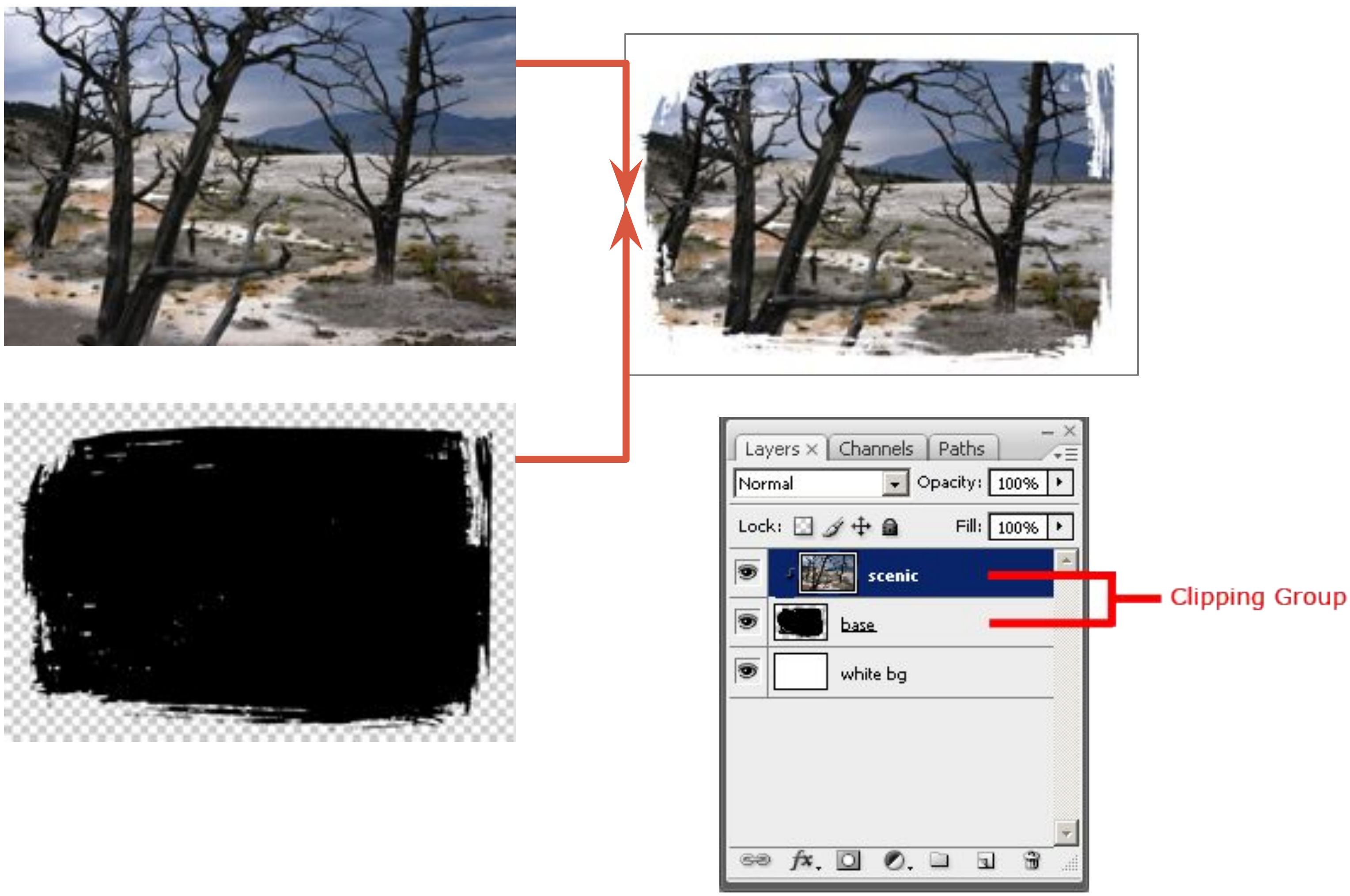
Layer Basics

* Stacking order of layers
* Reordering layers
* Opacity
* Blending mode
* Create new layer
* Delete layer
* Rename layer

Layer: Beyond Basics

* Layer style (e.g. drop shadow, bevel effects)
  * What is drop shadow http:/www.youtube.com/watch?v=5vcqPuTHVm8
  * What is bevel effects (http://www.rw-designer.com/bevel-effect-explained)
* Adjustment layers
* Layer mask
* Clipping mask

Clipping Mask Example



Thanks